SMARTCARE – CLINIC MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

UMAR FAROOK K

(2116240701573)

in partial fulfilment of the End Semester Examination for the course

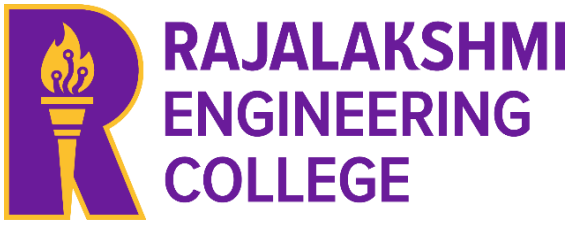
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

In

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



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(An Autonomous Institution, Affiliated to Anna University, Chennai)

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "SMARTCARE – CLINIC MANAGEMENT SYSTEM" is the bonafide work of **UMAR FAROOK K** (240701573), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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Submitted for the End Semester Examination (Individual Project) held on 11-11-2026.

INTERNAL EXAMINER	**EXTERNAL EXAMINER**

DECLARATION

I, UMAR FAROOK K (240701573), hereby declare that the project report titled "SMARTCARE – CLINIC MANAGEMENT SYSTEM", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 11-11-2026

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I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

UMAR FAROOK K

ABSTRACT

Small and mid-sized clinics still run on paper registers and loosely maintained spreadsheets. Patient details, appointment bookings, consultation notes, invoices, staff records and leave requests live in separate files, patient identifiers are assigned by hand, duplicate records creep in, and the administrator has no single view of how the clinic is performing. This project addresses that problem with SmartCare, a clinic management application built on the Zoho Creator low-code platform, whose rapid development, built-in hosting, role-based security and mobile access make it well suited to a clinic with no dedicated development team.

SmartCare comprises six forms — Patient Profiles, Appointment Scheduling, Medical Records, Billing Invoices, Staff Directory and Apply Leave — with 59 fields of 13 different types, ten reports (six list reports, a kanban board, an appointment calendar, a pivot chart and a pivot table) and a dashboard page with patient and doctor counters and distribution charts. Deluge scripts generate a unique patient code on form load and block duplicate patient codes on validation. Workflows send an e-mail notification for every new appointment and a follow-up reminder for medical records, a one-level approval process handles staff leave, and a seven-stage blueprint governs the appointment lifecycle from scheduling to consultation, cancellation or no-show. Access is controlled through six roles and a set of permission profiles for administrators, doctors, patients and other users, and the application has tablet and phone layouts.

The application was verified with a test plan that maps every functional requirement to test cases, and its design limitations are documented honestly, including the use of code-based pick lists in place of lookup fields. SmartCare replaces manual coding of patient identifiers, duplicate checking, status tracking and report compilation with automated, auditable workflows, and gives clinic staff a single, secure, mobile-ready source of truth.

PROJECT AT A GLANCE

Attribute	Details
Project Title	SmartCare – Clinic Management System
Application Domain	HealthCare
Student Name & Register No.	UMAR FAROOK K – 240701573
Zoho Creator Edition	Education
Application Name (Link Name)	SmartCare – Clinic Management System
Published / Live URL	https://creatorapp.zoho.in/umarfarook.k.2028.cse/smartcare-clinic-management-system
Demo Credentials (per role)	Admin: admin@123 | Patient: patient@123 | Doctor: doctor@123
Number of Forms / Subforms	6 / 2
Number of Reports (by type)	List: 6 Kanban: 1 Calendar: 1 Pivot/Chart: 2 (1 pivot chart, 1 pivot table) Map: 1 Others: 0 Total: 11
Number of Pages / Dashboards	1 (Dashboard)
Number of Workflows	Form: 2 (plus 1 form-properties e-mail notification) Schedule: 1 Approval: 1 Blueprint: 1 Batch: 0
Deluge Functions / Custom Actions	2 custom Deluge scripts (form workflows); 4 sendmail tasks; 0 stand-alone functions; 0 custom actions
Integrations	Zoho Mail notifications (sendmail); address field with map link. No external API or Zoho CRM integration.
Roles & Profiles Defined	6 roles: CEO, Admin, Receptionist, Doctor, Patient, Pharmacist. Profiles: Administrator, Read, Write, Developer, Doctor, Admin, Patient, plus customer-portal profiles Admin and Customer
Environments Used	Zoho Creator
Date of Production Deployment	10-11-2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CO	Course Outcome
CRM	Customer Relationship Management
DS	Deluge Script
ePHI	Electronic Protected Health Information
ER	Entity–Relationship
FR	Functional Requirement
IEEE	Institute of Electrical and Electronics Engineers
INR	Indian Rupee
NFR	Non-Functional Requirement
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
REST	Representational State Transfer
SMS	Short Message Service
TC	Test Case
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
Zia	Zoho's AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Outpatient clinics are among the most process-heavy small organisations. A single patient visit touches the front desk (registration and appointment booking), the doctor (consultation, diagnosis and prescription), the billing counter (invoice and insurance claim) and, behind the scenes, the administrator (staff roster, leave and reports). The clinic that SmartCare is designed for is a multi-department outpatient clinic with General, Cardiology, Orthopedics, Emergency and Pharmacy departments, staffed by receptionists, doctors, nurses, pharmacists and an administrator.

In such a clinic the workflow is traditionally paper-based: appointments are noted in a register, patient files are kept in folders, prescriptions are handwritten and bills are prepared separately. Each of these records is a silo, and combining them to answer a simple question — how many patients are registered, which appointments are pending, who has an outstanding invoice — requires manual compilation. SmartCare brings these functions into one application.

1.2 Problem Statement

A clinic needs a single, secure and easy-to-maintain system to register patients, schedule and track appointments, record consultations, raise invoices and manage staff and leave, but it has neither the budget nor the staff for a conventional software project. The absence of such a system leads to the following pain points:

- **Duplicated and inconsistent data:** the same patient is recorded in several registers or spreadsheets, and patient codes are assigned manually, which causes duplicates.
- **No visibility of appointment status:** there is no controlled lifecycle (scheduled, confirmed, checked-in, in consultation, completed, cancelled, no-show), so the status of a visit is known only to whoever last handled it.
- **Delayed and untracked approvals:** staff leave is requested informally and the outcome is not communicated or recorded systematically.
- **Manual reporting:** the administrator compiles counts and summaries by hand, such as the number of patients, blood-group distribution and invoice payment status.

- **Weak access control for sensitive data:** paper files and shared spreadsheets give every user access to medical and personal information.

1.3 Objectives

- To design a data model and forms for patient profiles, appointments, medical records, billing invoices, staff directory and leave applications.
- To automate patient-code generation, duplicate prevention, e-mail notifications, leave approval and the appointment lifecycle using Deluge scripting, workflows, an approval process and a blueprint.
- To provide a dashboard and reports (list, kanban, calendar and pivot) that give clinic staff real-time visibility of patients, appointments, records and billing.
- To protect patient and staff data through roles, permission profiles and personal-data flags on sensitive fields.
- To make the application available on web, tablet and phone layouts and deploy it to a production environment with governance controls.

1.4 Scope of the Project

In scope: patient registration with demographic, contact, address, blood group, emergency contact and insurance details; appointment scheduling and lifecycle tracking; medical records with diagnosis, treatment plan, prescription and follow-up; billing invoices with payment status and insurance claim reference; staff directory; leave application and approval; a dashboard; role-based access for administrator, doctor, patient and other roles; and web, tablet and phone layouts.

Out of scope: payment-gateway processing, integration with laboratory or pharmacy-inventory systems, inventory management, SMS notifications, e-prescription signing, and any pro-code extension. These are listed as future enhancements in Chapter 9.

1.5 Justification for a Low-Code Approach

Low-code platforms let developers and trained business users build applications largely through visual configuration, with hand-written code reserved for the logic that needs it [1], [2]. For SmartCare this approach is justified on five grounds. First, **development speed**: forms, reports, pages, approvals and blueprints are configured rather than coded, and Zoho Creator generates the database, the user interface and the application programming interface automatically [3]. Second, **maintainability by a citizen**

developer: a clinic administrator can add a field or change a status list without a release cycle. Third, **built-in hosting and mobile access**: no server has to be provisioned, and the same application is available on web, tablet and phone. Fourth, **built-in security**: roles, permission profiles and personal-data flags are platform features. Fifth, **cost**: the clinic pays for a platform subscription rather than for a custom development team and infrastructure.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing system, a patient arrives at the clinic or telephones for an appointment, and the receptionist notes the booking in a paper register or a spreadsheet. A paper file is created for a new patient and a code is assigned by hand. After consultation the doctor writes the diagnosis and prescription on paper, the bill is prepared separately, and staff leave is requested on paper or by message. At the end of the period, the administrator compiles the registers by hand to prepare reports. Figure 2.1 shows this as-is flow.

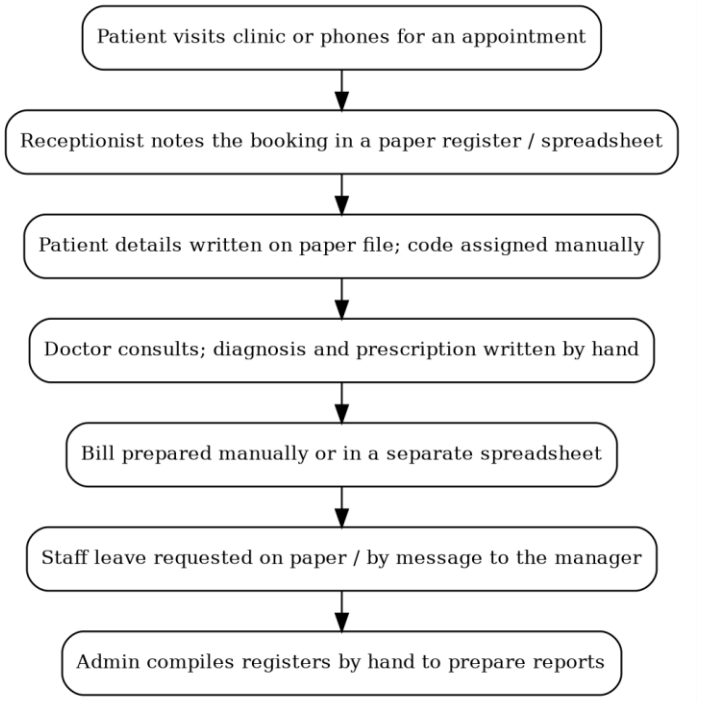


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Patient codes are assigned manually, so duplicate or inconsistent codes are likely and nothing checks them.
- Appointments, consultations and invoices are held in separate records and cannot be cross-referenced quickly.
- There is no controlled status for an appointment and no automatic notification when a new booking is made.
- Leave approval is informal and its outcome is not recorded or communicated.

- Every person with access to the paper files or shared spreadsheet can see all patient and salary data.
- Reports must be compiled by hand and are therefore late and error-prone.

2.3 Low-Code versus Traditional Development

Table 2.1 compares the two approaches for SmartCare. The effort figures are indicative estimates for an application of this size (six forms, ten reports, one page, five workflows) and are not measured values; the Deluge language used in the low-code build is documented in [4].

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Several weeks: schema design, back-end APIs, front-end screens, authentication	A few days to two weeks: forms, reports and workflows configured in the builder
Skills required	Back-end, front-end, database and DevOps skills	Data-modelling knowledge and basic Deluge scripting
Hosting & infrastructure	Server, database and deployment pipeline to be provisioned and maintained	Hosted by Zoho; no infrastructure to manage
Mobile application	Separate native or responsive build	Phone and tablet layouts and the Creator mobile app are generated from the same application
Security & access control	Authentication and role logic to be coded and tested	Roles, permission profiles and personal-data flags are built in
Maintenance & change	Code change, test and redeploy for each new field or status	Edit in the builder; changes can be published through environments
Customisation limits	Practically unlimited	Bounded by platform features and Deluge capabilities
Cost	High up-front development and hosting cost	Subscription-based; low up-front cost

2.4 Literature Survey

Table 2.2 summarises the sources reviewed for this project: research on low-code platforms, analyst and vendor documentation, and an existing clinic-management product. Full citations are given in the References.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	A. C. Bock and U. Frank, 2021 [2]	Low-Code Platform	Defines low-code platforms and positions them within model-driven development, emphasising visual, declarative application construction.	Provides the conceptual basis for building SmartCare through declarative forms, reports and workflows.
2	Y. Luo et al., 2021 [5]	Characteristics and Challenges of Low-Code Development: The Practitioners' Perspective	Practitioner discussions highlight benefits such as speed and ease of use, and challenges such as limited customisation and platform lock-in.	Informs the limitations discussed in Section 8.4.
3	D. Di Ruscio et al., 2022 [6]	Low-code development and model-driven engineering: Two sides of the same coin?	Compares low-code with model-driven engineering and identifies shared concepts and open challenges.	Supports the data-model-first design used for the six forms.
4	K. Roķis and M. Kirikova, 2022 [7]	Challenges of Low-Code/No-Code Software Development: A Literature Review	Identifies challenges in the low-code development process and suggests ways to mitigate them.	Guides the testing, governance and limitation analysis.
5	K. Roķis and M. Kirikova, 2023 [8]	Exploring Low-Code Development: A Comprehensive Literature Review	Summarises definitions, tools, lifecycles, application areas, benefits and challenges of low-code development.	Supports the justification for a low-code approach (Section 1.5).
6	Zoho Corporation [3]	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows, blueprints, approvals and roles in Zoho Creator.	Primary technical reference for every component built.
7	Zoho Corporation [4]	Deluge Script Reference Guide	Documents Deluge syntax, tasks and built-in functions.	Reference for DS-01 to DS-06.
8	Practo [9]	Practo Ray (clinic management software)	Commercial cloud product offering appointments, patient records and billing for clinics.	Existing product in the same domain; SmartCare delivers a comparable core on a low-code platform the clinic can modify itself.

2.5 Emerging Trends in Low-Code Development

Four trends shape the low-code field [7], [8], [10]. **Citizen development** allows domain experts to build and maintain applications; SmartCare's configuration-first design supports this, since statuses, departments and roles are pick lists an administrator can edit. **AI-assisted building** is now offered by many platforms, including Zoho Creator, to generate applications and scripts from prompts; SmartCare does not use it in this version, and Section 5.9 explains how it could be applied. **Hyper-automation** combines workflows, approvals and notifications; SmartCare applies this on a small scale through its blueprint, approval process and e-mail notifications. **Composable**

applications assemble business capabilities from reusable parts; the six independent forms and their reports are designed so that new modules, such as pharmacy inventory, can be added without redesign.

2.6 Proposed System

The proposed system, SmartCare, replaces the paper and spreadsheet processes with a Zoho Creator application. Patient registration generates a unique patient code automatically and rejects duplicates, which addresses the first limitation in Section 2.2. All six forms store their data in one platform, and their reports and dashboard give a combined view, which addresses the siloed-records and manual-reporting limitations. The appointment blueprint gives each visit a controlled status, and a workflow notifies the administrator when a booking is added. Leave requests pass through an approval process and the outcome is e-mailed to the applicant. Roles and permission profiles restrict who can view, edit or delete data.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator (Admin / CEO)	Clinic owner or manager with full control	Manage staff directory, approve leave, manage users and profiles, view dashboard and all reports, delete records
Receptionist	Front-desk staff	Register patients, schedule appointments and move them through the blueprint, raise invoices, apply for leave
Doctor	Consulting physician	View and edit appointments, patient profiles and medical records; record diagnosis, treatment plan and prescription
Pharmacist / Nurse	Clinical support staff	View medical records and prescriptions, apply for leave
Patient (portal user)	Person receiving care	View permitted reports and add appointment details through the customer-portal profile

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow staff to register a patient with name, date of birth, gender, phone, e-mail, address, blood group, emergency contact, insurance provider and medical history summary.	Receptionist, Admin	High
FR-02	The system shall automatically generate a unique patient code when a new patient form is opened.	System	High
FR-03	The system shall reject a patient record whose patient code already exists.	System	High
FR-04	The system shall allow an appointment to be scheduled with code, date, time, reason, status, patient, doctor and notes.	Receptionist, Patient	High
FR-05	The system shall control the appointment lifecycle through defined stages and transitions (confirm, check in, start and complete consultation, cancel, mark no-show).	Receptionist, Doctor	High
FR-06	The system shall e-mail the administrator when a new appointment entry is added.	System	Medium
FR-07	The system shall record diagnosis, treatment plan, prescription, visit type, follow-up requirement and next visit date for each consultation.	Doctor	High
FR-08	The system shall send a follow-up reminder e-mail for a medical record on its scheduled date.	System	Medium
FR-09	The system shall maintain a staff directory with role, department, specialization, hire date, salary and status.	Admin	High

FR ID	Requirement	Role(s)	Priority
FR-10	The system shall record invoices with total, paid and balance amounts in rupees, payment status and insurance claim reference.	Receptionist, Admin	High
FR-11	The system shall let staff apply for leave, route the request for approval and e-mail the outcome to the applicant.	Staff, Admin	High
FR-12	The system shall provide list reports for every form, a kanban board of medical records by visit type and a calendar of appointments.	All	Medium
FR-13	The system shall provide a pivot chart of appointment status and a pivot table of invoice payment summary.	Admin	Medium
FR-14	The system shall display a dashboard with total patients, total doctors, blood-group distribution, role distribution and a button to add an appointment.	Admin, Doctor	Medium
FR-15	The system shall restrict access to forms and reports by role and permission profile (view, edit, delete).	All	High
FR-16	The system shall be usable in web, tablet and phone layouts.	All	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports with up to 1,000 records shall load within 3 seconds on a broadband connection.
NFR-02	Security	Users shall access only the forms and reports permitted by their profile; sensitive fields shall be flagged as personal data.
NFR-03	Usability	The application shall be usable on web, tablet and phone layouts and on the Zoho Creator mobile app.
NFR-04	Availability	The application shall be available through the Zoho Creator cloud service without any self-managed server.
NFR-05	Maintainability	Lists such as status, department and role shall be editable through the builder without code changes.
NFR-06	Auditability	Record changes shall be traceable through the Zoho Creator audit trail.
NFR-07	Localisation	Dates shall display as dd-MMM-yyyy in the Asia/Kolkata time zone with a 24-hour clock; amounts shall be in rupees with Indian digit grouping; phone numbers shall be restricted to India.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator **[ENTER EDITION]** Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)

Category	Specification
Other Zoho services	Zoho Mail (used by sendmail notifications)
Mobile	Zoho Creator mobile app (Android / iOS) and the phone and tablet layouts of the application
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

Figure 3.1 shows the actors and use cases of SmartCare. Tables 3.5 to 3.7 describe three key use cases in detail.

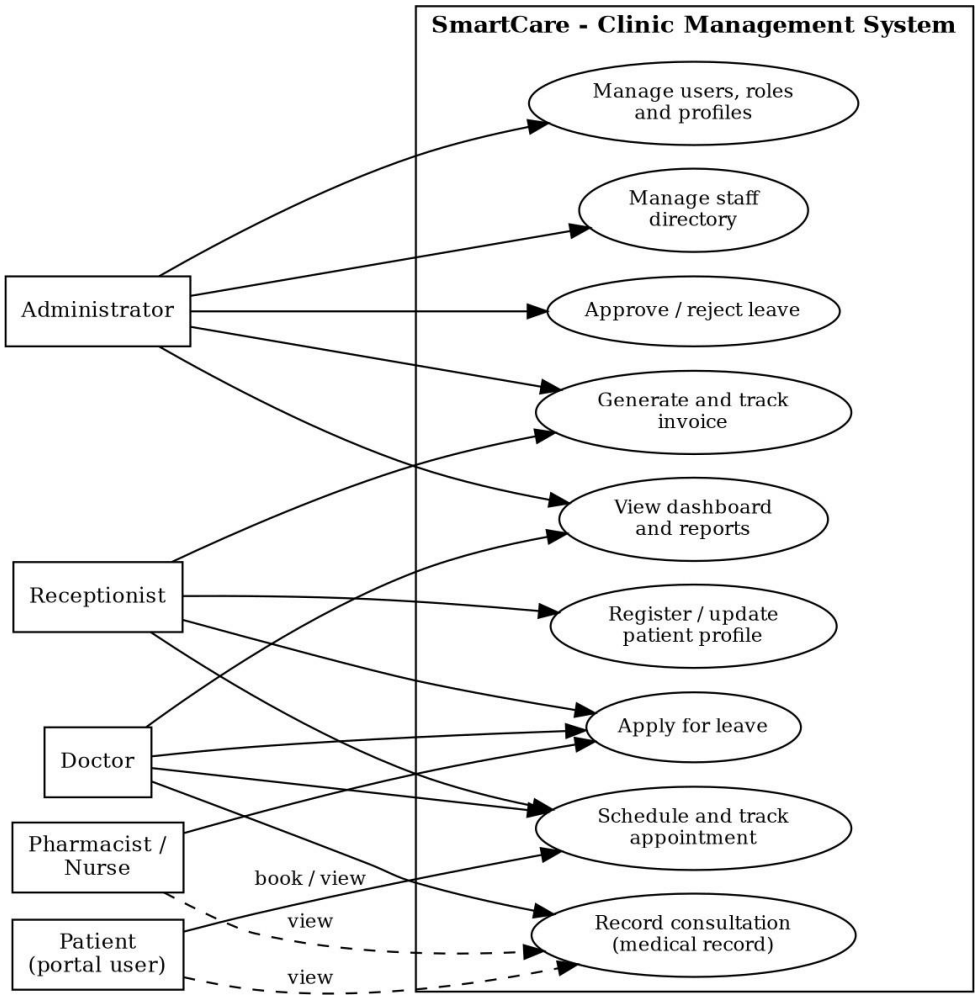


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Register Patient

Use Case ID	UC-01
Use Case Name	Register Patient
Actor(s)	Receptionist, Administrator
Pre-condition	User is logged in with permission to add Patient Profiles.

Main Flow	1. User opens the Patient Profiles form. 2. The system generates a patient code (DS-01). 3. User enters name, date of birth, gender, phone, e-mail, address, blood group, emergency contact and insurance details. 4. User clicks Submit. 5. The system validates the patient code (DS-02) and saves the record.
Alternate Flow	If the patient code already exists, the system shows an alert and cancels the submission; the user corrects the code and resubmits.
Post-condition	A new patient record exists and appears in the All Patient Profiles report and the dashboard counter.
Related FR	FR-01, FR-02, FR-03

Table 3.6: Use Case Description – Schedule and Progress an Appointment

Use Case ID	UC-02
Use Case Name	Schedule and Progress an Appointment
Actor(s)	Receptionist, Doctor
Pre-condition	The patient and the doctor exist in the system.
Main Flow	1. Receptionist opens the Appointment Scheduling form and enters code, date, time, reason, patient, doctor and notes. 2. The system saves the record at the stage “Appointment Scheduled” and e-mails the administrator. 3. Staff move the record through Confirm Appointment, Check In Patient, Start Consultation and Complete Consultation.
Alternate Flow	At “Scheduled” or “Confirmed” the user may use Cancel Appointment; at “Confirmed” the user may use Mark Patient No Show.
Post-condition	The appointment is in a terminal stage (Completed, Cancelled or No Show) and appears in the calendar and the pivot chart.
Related FR	FR-04, FR-05, FR-06, FR-12, FR-13

Table 3.7: Use Case Description – Apply for and Approve Leave

Use Case ID	UC-03
Use Case Name	Apply for and Approve Leave
Actor(s)	Staff member, Administrator
Pre-condition	Staff member has access to the Apply Leave form.
Main Flow	1. Staff member completes all mandatory fields (staff code, name, role, leave type, from, to, number of days, reason) and clicks Apply. 2. The request enters the Leave Approval process. 3. The approver approves or rejects the request. 4. The system e-mails the outcome to the applicant.
Alternate Flow	If any mandatory field is empty the form is not submitted.
Post-condition	The leave record carries its approval status and the applicant has been notified.
Related FR	FR-11

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

SmartCare follows the four-layer architecture of a Zoho Creator solution, shown in Figure 4.1. Users of five kinds reach the application through a web browser, a tablet layout or the phone layout of the Creator mobile app (layer 1). The application layer (layer 2) holds the six forms, ten reports, the dashboard page and the automation — Deluge form workflows, a schedule, an approval process and a blueprint. All form data is held in the Zoho Creator data store (layer 3), where sensitive fields carry personal-data flags. The platform services layer (layer 4) supplies Zoho Mail for notifications, role and profile security, and map support for the address field. No external application programming interface is called.

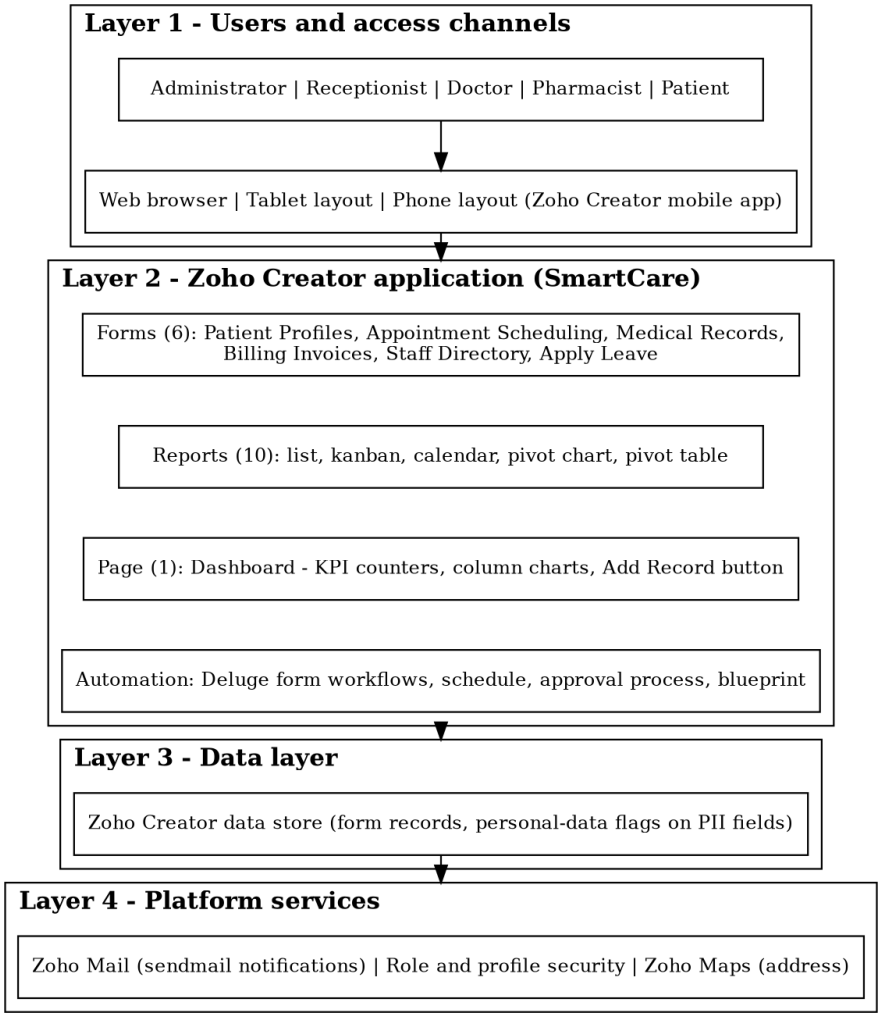


Figure 4.1: System Architecture of the Application

4.2 Data Model

The data model has six entities, shown in Figure 4.2. Patient_Profiles and Staff_Directory are the master entities. Appointments, medical records and invoices each refer to one patient and one doctor or service provider, giving the one-to-many relationships in Table 4.1. Apply_Leave refers to a staff member through the staff code.

In the current implementation these references are held as **pick-list fields of codes** (for example, patient codes P-1001 to P-1005 and staff codes STF001 to STF005) rather than Zoho Creator lookup fields, and no subforms are used. The relationships are therefore logical rather than enforced by the platform. This is documented as a limitation in Section 8.4 and as the first future enhancement in Section 9.2.

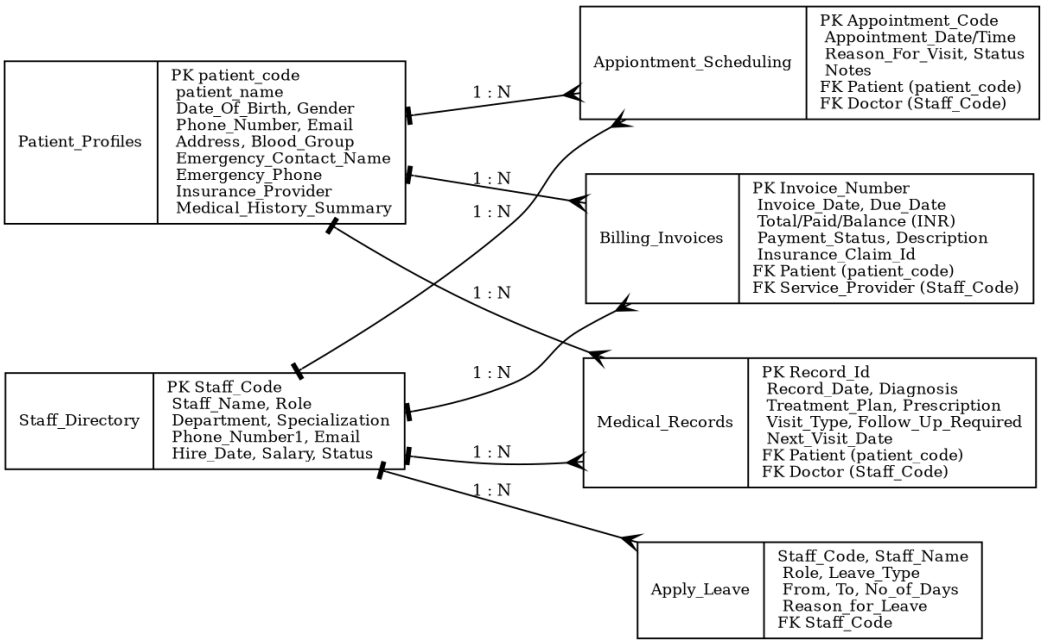


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Patient_Profiles	patient_code, patient_name, Date_Of_Birth, Blood_Group	Appiontment_Scheduling	One-to-Many	Pick list “Patient” (patient codes)
Patient_Profiles	patient_code	Medical_Records	One-to-Many	Pick list “Patient”
Patient_Profiles	patient_code	Billing_Invoices	One-to-Many	Pick list “Patient”
Staff_Directory	Staff_Code, Staff_Name, Role, Department	Appiontment_Scheduling	One-to-Many (doctor)	Pick list “Doctor” (staff codes)
Staff_Directory	Staff_Code	Medical_Records	One-to-Many (doctor)	Pick list “Doctor”

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Staff_Directory	Staff_Code	Billing_Invoices	One-to-Many (service provider)	Pick list "Service_Provider"
Staff_Directory	Staff_Code	Apply_Leave	One-to-Many	Text field "Staff_Code"

4.3 Form Design

Tables 4.2 to 4.7 give the field specification of each form, using Zoho Creator field-type names. The "Remarks" column records personal-data flags and other configuration. All forms show the buttons Submit and Reset on add, and Update and Cancel on edit (the Apply Leave form labels its add button "Apply").

Table 4.2: Field Specification – Patient Profiles Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Patient Name	patient_name	Name	No	First and last name shown; prefix and suffix hidden	Personal data
Patient Code	patient_code	Single Line	No	Auto-filled by DS-01; uniqueness checked by DS-02	System generated
Date Of Birth	Date_Of_Birth	Date	No	All days allowed	
Gender	Gender	Drop Down	No	Male, Female, Prefer Not To Say	
Phone Number	Phone_Number	Phone	No	India only; default country India	Personal data
Email	Email	Email	No	Maximum 80 characters	Personal data
Address	Address	Address	No	Captures coordinates; map adjustment off	Personal data
Blood Group	Blood_Group	Drop Down	No	A+, B+, AB+, A-, B-, Unknown	Used by dashboard chart
Emergency Contact	Emergency_Contact_Name	Name	No	First and last name	Personal data
Emergency Phone	Emergency_Phone	Phone	No	India only	Personal data
Insurance Provider	Insurance_Provider	Single Line	No		
Medical History Summary	Medical_History_Summary	Multi Line	No	Height 100 px	

Table 4.3: Field Specification – Appointment Scheduling Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Appointment Code	Appointment_Code	Single Line	No		Entered by user
Appointment Date	Appointment_Date	Date	No	All days allowed	Calendar start date
Appointment Time	Appointment_Time	Time	No	Format hh:mm:ss	
Reason For Visit	Reason_For_Visit	Single Line	No		
Status	Status	Drop Down	No	Scheduled, Completed, Cancelled, Other	Used by pivot chart
Patient	Patient	Drop Down	No	Patient codes P-1001 to P-1005	Code-based pick list
Doctor	Doctor	Drop Down	No	Staff codes STF001 to STF005	Code-based pick list
Notes	Notes	Multi Line	No	Height 100 px	
(Blueprint stage)	—	Blueprint	—	Starts at "Appointment Scheduled"	See Section 5.6

Table 4.4: Field Specification – Medical Records Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Record Id	Record_Id	Single Line	No		
Record Date	Record_Date	Date-Time	No	Format hh:mm:ss	Schedule start field
Diagnosis	Diagnosis	Single Line	No		
Treatment Plan	Treatment_Plan	Multi Line	No	Height 100 px	
Prescription	Prescription	Multi Line	No	Height 100 px	
Doctor	Doctor	Drop Down	No	Staff codes STF001 to STF005	Code-based pick list
Patient	Patient	Drop Down	No	Patient codes P-1001 to P-1005	Code-based pick list
Visit Type	Visit_Type	Drop Down	No	Checkup, Inpatient, Outpatient, Emergency, Other	Kanban display field
Follow Up Required	Follow_Up_Required	Checkbox	No	Single option "Follow Up Required"	
Next Visit Date	Next_Visit_Date	Date	No	All days allowed	

Table 4.5: Field Specification – Billing Invoices Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Invoice Number	Invoice_Number	Number	No		

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Invoice Date	Invoice_Date	Date	No		
Due Date	Due_Date	Date	No		
Total Amount	Total_Amount	Currency (INR)	No	Indian comma-dot format	
Paid Amount	Paid_Amount	Currency (INR)	No	Indian comma-dot format	
Balance Due	Balance_Due	Currency (INR)	No	Indian comma-dot format	Entered, not a formula
Payment Status	Payment_Status	Drop Down	No	Partial, Paid, Overdue, Pending, Other	Used by pivot table
Patient	Patient	Drop Down	No	Patient codes P-1001 to P-1005	Code-based pick list
Service Provider	Service_Provider	Drop Down	No	Staff codes STF001 to STF005	Code-based pick list
Description	Description	Multi Line	No	Height 100 px	
Insurance Claim Id	Insurance_Claim_Id	Single Line	No		

Table 4.6: Field Specification – Staff Directory Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Staff Name	Staff_Name	Name	No	First name shown; last name, prefix and suffix hidden	Personal data
Staff Code	Staff_Code	Single Line	No		
Role	Role	Drop Down	No	Receptionist, Pharmacist, Doctor, Nurse, Admin, Other	Dashboard counter and chart
Department	Department	Drop Down	No	Orthopedics, Pharmacy, General, Emergency, Cardiology, Other	
Phone Number	Phone_Number1	Phone	No	India only	Personal data
Email	Email	Email	No	Maximum 80 characters	Personal data
Specialization	Spealization	Single Line	No		Link name misspelt; label correct
Hire Date	Hire_Date	Date	No		
Salary	Salary	Currency (INR)	No	Indian comma-dot format	Sensitive
Status	Status	Drop Down	No	Active, On Leave, Inactive, Other	

Table 4.7: Field Specification – Apply Leave Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Staff Code	Staff_Code	Single Line	Yes		
Staff Name	Staff_Name	Name	Yes	First and last name; prefix and suffix hidden	Personal data
Role	Role	Drop Down	Yes	Receptionist, Pharmacist, Doctor, Nurse, Other	
Leave Type	Leave_Type	Drop Down	Yes	Casual Leave, Maternity Leave, Sick Leave, Other	
From	From	Date	Yes	All days allowed	
To	To	Date	Yes	All days allowed	
No of Days	No_of_Days	Number	Yes		
Reason for Leave	Reason_for_Leave	Multi Line	Yes	Height 100 px	

4.4 Process and Workflow Design

Three processes are automated. **Patient registration** (Figure 4.3) uses two Deluge form workflows: DS-01 fills the patient code when the form loads and DS-02 rejects a duplicate on validation. The **appointment lifecycle** (Figure 4.4) is a blueprint with seven stages, starting at “Appointment Scheduled”, and transitions that offer only the valid next actions. **Leave approval** (Figure 4.5) is a one-level approval process whose approve and reject actions e-mail the applicant.

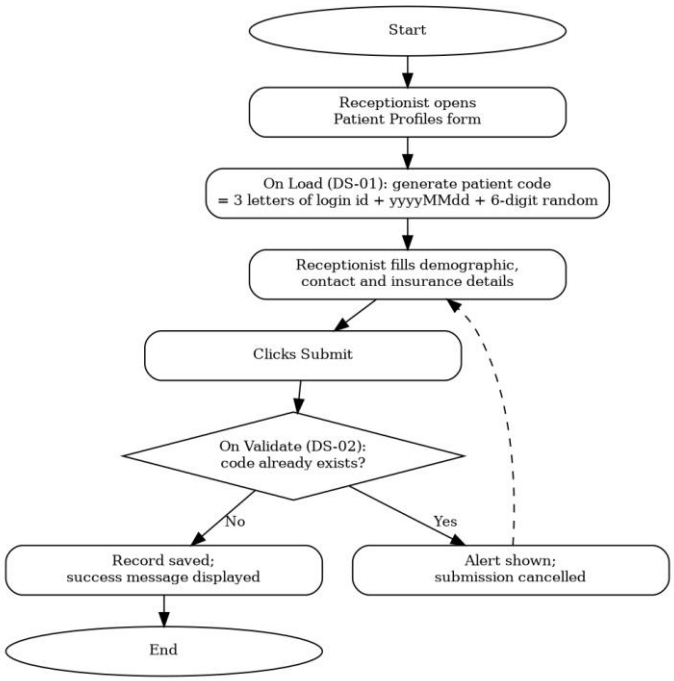


Figure 4.3: Process Flow for Patient Registration

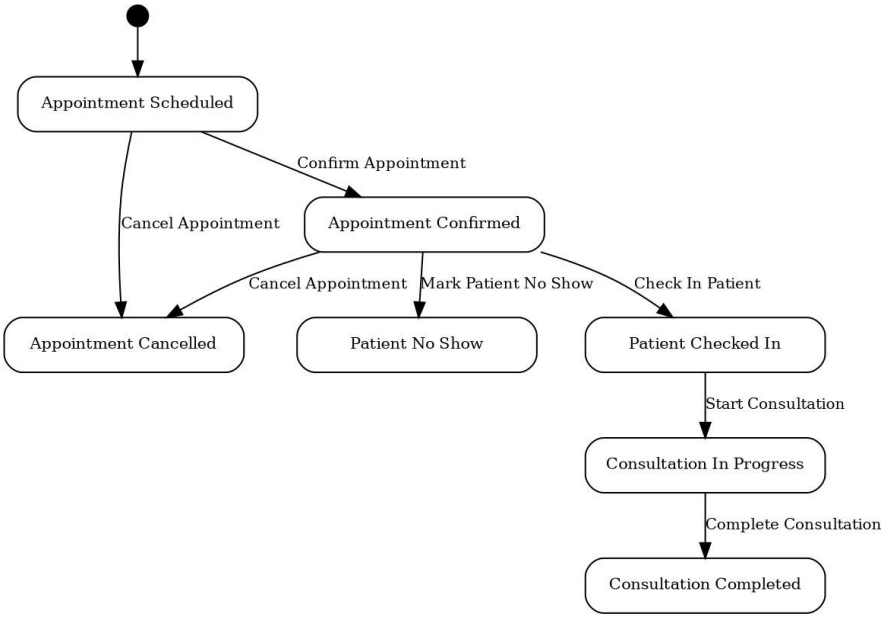


Figure 4.4: Blueprint Flow for the Appointment Lifecycle

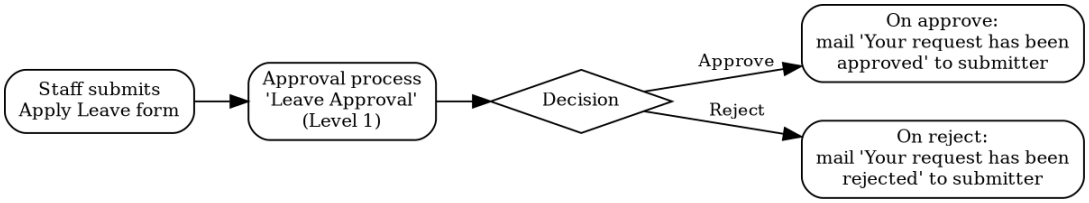


Figure 4.5: Process Flow for Leave Approval

4.5 User Interface and Navigation Design

The left-hand menu is organised into sections, one per business module, each holding the form and its reports (Figure 4.6). Dashboard, Leaves, Approvals (pending and completed tasks) and App Preferences sit alongside the module sections. The application uses the Lato font with an outline icon style, an "SCM" application icon, labels placed to the left of fields on the web layout and automatic label placement on phone and tablet.

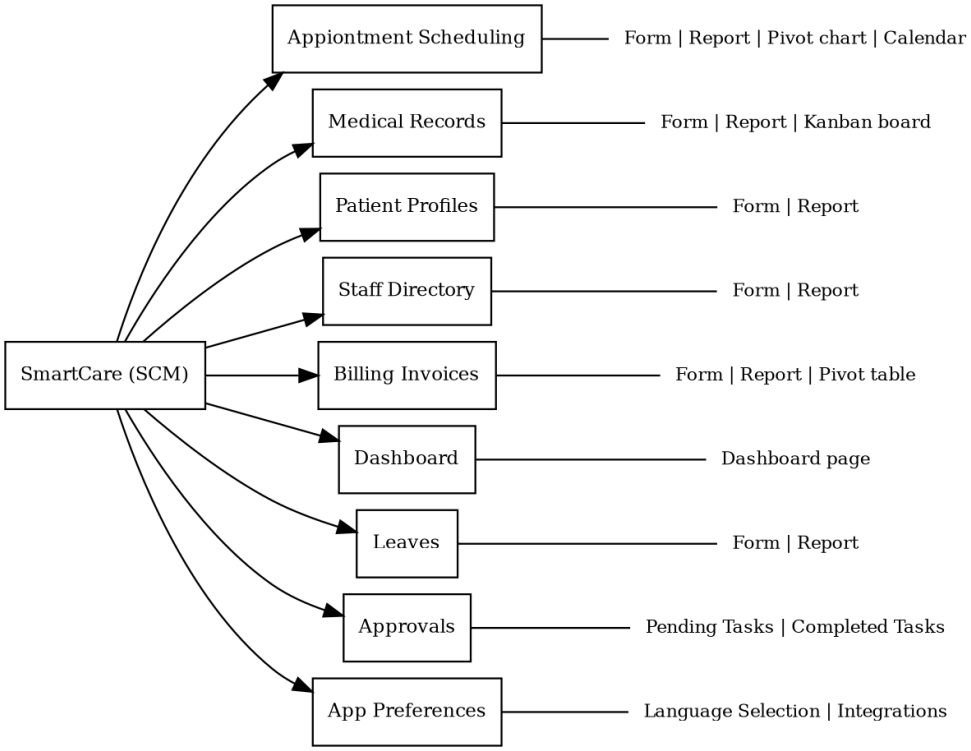


Figure 4.6: Navigation Structure of the Application

4.6 Access Control Design

Access is designed around permission profiles, listed in Table 4.8 with A = Add, E = Edit, V = View and D = Delete. The Administrator profile has all permissions; the Admin, Doctor and Patient profiles define module-level and report-level permissions. A "—" marks a component that the profile's exported definition does not grant.

Table 4.8: Role–Permission Matrix

Component (form and its list report)	Admin profile	Doctor profile	Patient profile
Patient Profiles	A E V D	A E V	A V
Appiontment Scheduling	A E V D	A E V	A V
Medical Records	A E V D	A E V	A V
Billing Invoices	A E V D	A E V	A V

Component (form and its list report)	Admin profile	Doctor profile	Patient profile
Staff Directory	A E V D	A E V	A V
Dashboard page	V	V	V
Apply Leave, Kanban board, Calendar	—	—	—

The Administrator, Read and Write profiles apply across all components, and the customer-portal profiles (Admin and Customer) extend the same model to portal users. Section 8.4 and Section 9.2 note that record-level criteria, so that a patient sees only their own records, are not yet configured.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application “SmartCare - Clinic Management System” was created in the Zoho Creator workspace of the developer account **from scratch**. Its regional settings are the date format dd-MMM-yyyy, the Asia/Kolkata time zone and a 24-hour clock. The menu has ten sections: six module sections (Appiontment Scheduling, Medical Records, Patient Profiles, Staff Directory, Billing Invoices and Leaves), the Dashboard section, an Approvals section with Pending Tasks and Completed Tasks, a Reports section for shared analytics, and the App Preferences section with Language Selection and Integrations. The theme uses the Lato font, an outline icon style and an application icon with the text “SCM”.

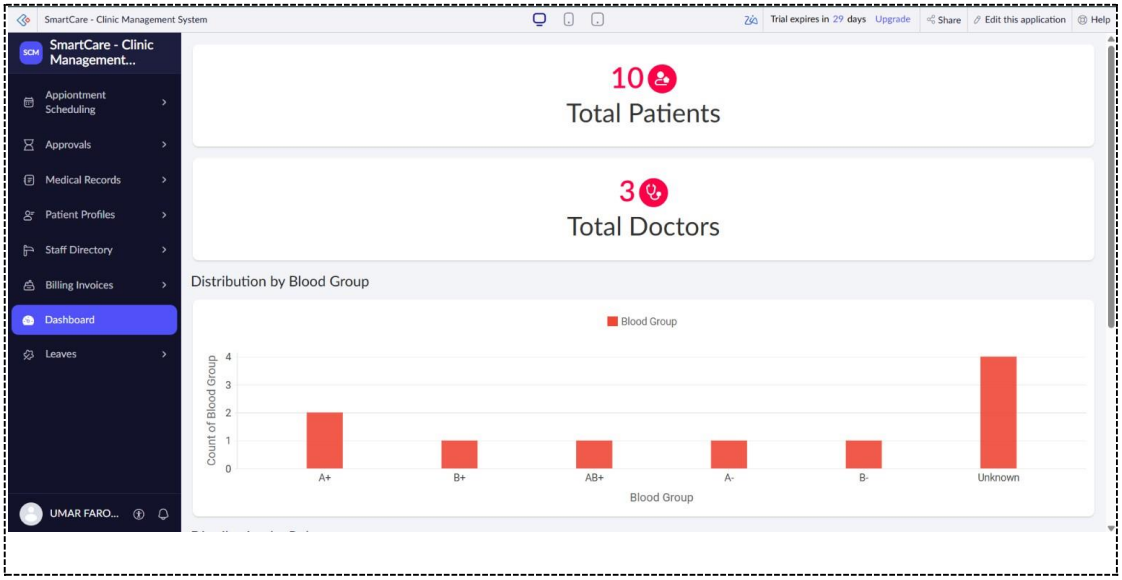


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

The six forms capture 59 fields of 13 different Zoho Creator field types: Single Line, Multi Line, Date, Time, Date-Time, Drop Down, Number, Currency, Name, Phone, Email, Address and Checkbox. Field-level configuration includes the following:

- **Mandatory fields:** every field of the Apply Leave form is mandatory, so a leave request cannot be submitted incomplete.

- **Input restriction:** phone fields accept Indian numbers only with India as the default country; e-mail fields are limited to 80 characters; amounts use the rupee currency type with Indian digit grouping.
- **Compound fields:** Name fields show first and last name with prefix and suffix hidden; the Address field captures map coordinates and the reports show a link to the map.
- **Personal-data flags:** patient name, phone, e-mail, address, emergency contact and phone, and staff name, phone and e-mail are flagged as personal data.
- **Form properties:** each form has its own success message (for example "Patient Profile Added Successfully") and the Submit / Reset and Update / Cancel buttons.

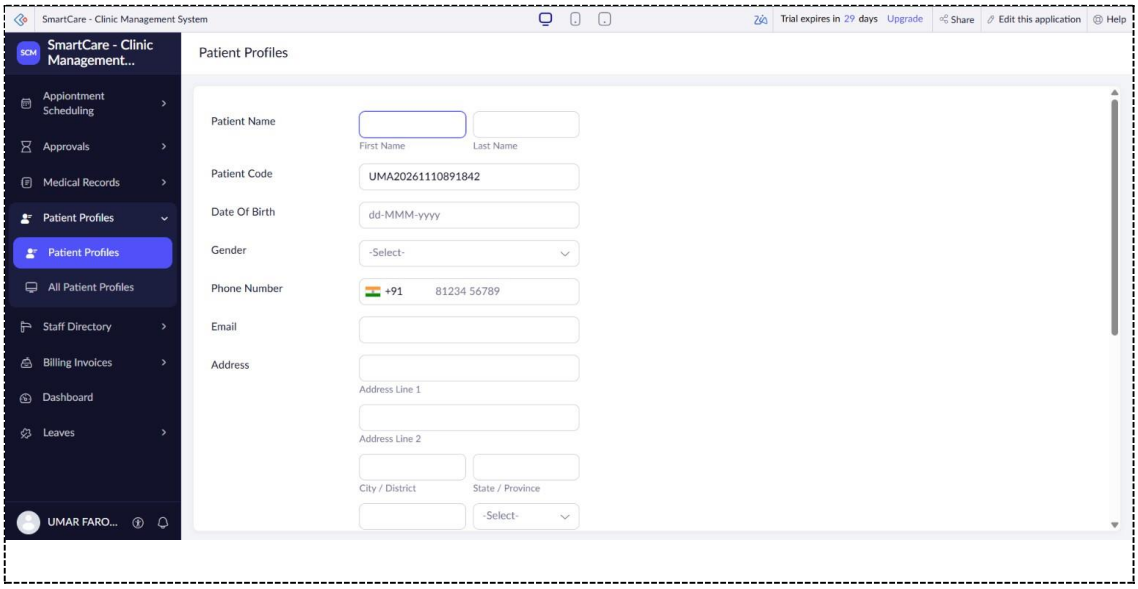


Figure 5.2: Patient Profiles Form in the Form Builder

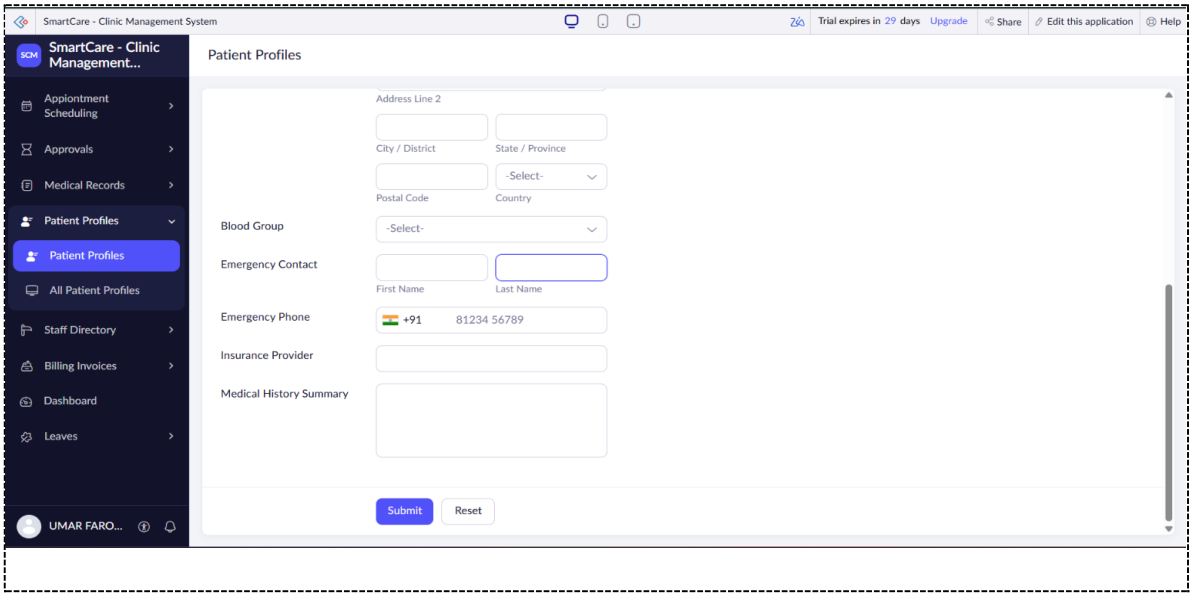


Figure 5.3: Patient Profiles Form in Live Mode

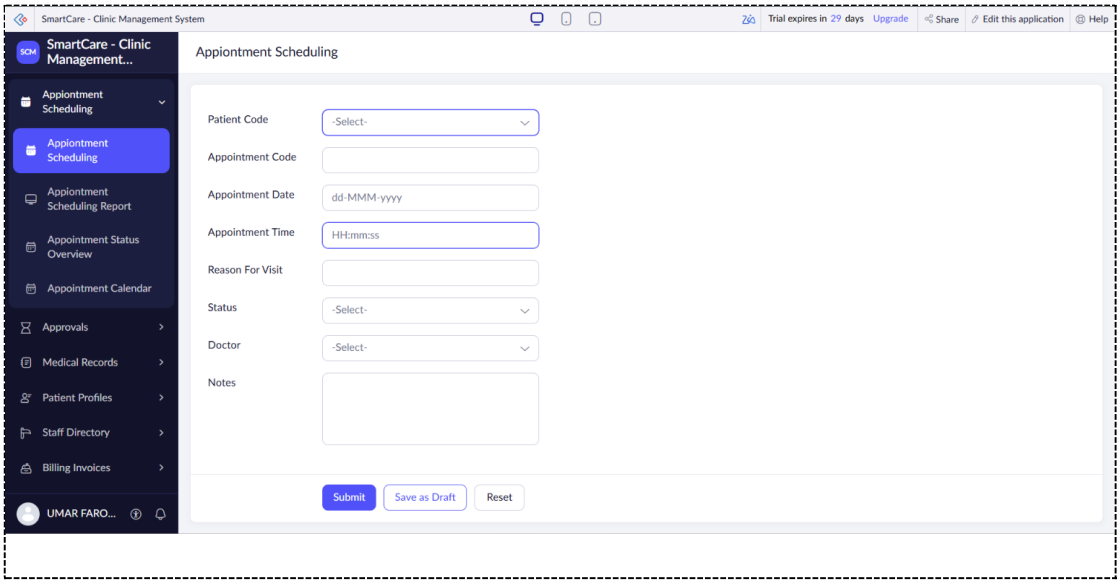


Figure 5.4: Appiontment Scheduling Form in Live Mode

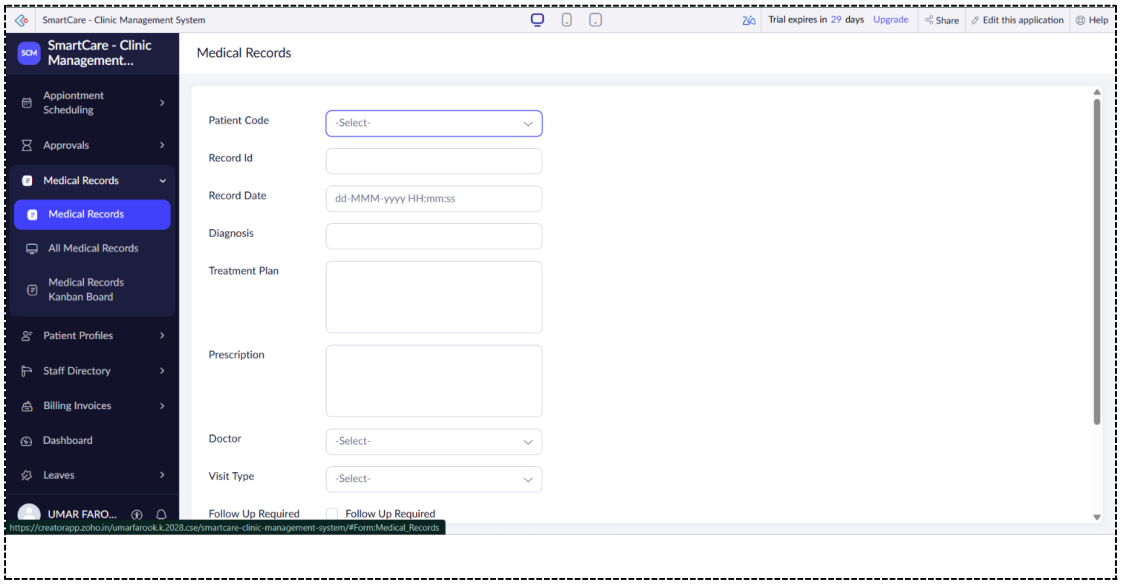


Figure 5.5: Medical Records, Billing Invoices, Staff Directory and Apply Leave Forms in Live Mode

5.3 Reports

Ten reports were built, as listed in Table 5.1: six list reports (one per form), a kanban board, a calendar and two analytical reports. List reports show the main fields of their form with friendly column headings; the Address column offers a link to the map and phone columns use the plain-number display format. Each list report has quick-view and detail layouts and header and record menus offering Edit, Duplicate and Delete.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Grouping / Display	Custom Actions	Roles
Appiontment_Scheduling_Report	List	Appiontment_Scheduling	All rows; 8 columns	Edit, Duplicate, Delete menu	Admin, Doctor, Patient
All_Patient_Profiles	List	Patient_Profiles	All rows; 12 columns; address map link	Edit, Duplicate, Delete menu	Admin, Doctor, Patient
All_Medical_Records	List	Medical_Records	All rows; 10 columns	Edit, Duplicate, Delete menu	Admin, Doctor, Patient
All_Billing_Invoices	List	Billing_Invoices	All rows; 11 columns	Edit, Duplicate, Delete menu	Admin, Doctor, Patient
Staff_Directory_Report	List	Staff_Directory	All rows; 10 columns	Edit, Duplicate, Delete menu	Admin, Doctor, Patient

Report Name	Type	Source Form	Grouping / Display	Custom Actions	Roles
All_Leaves	List	Apply_Leave	All rows; 8 columns	Edit, Duplicate, Delete menu	Portal Admin profile
Medical_Records_Kanban_Board	Kanban	Medical_Records	Cards displayed by Visit_Type	—	Portal Admin profile
Appointment_Calendar	Calendar	Appiontment_Scheduling	Start date Appointment_Date; label Appointment_Code	—	Portal Admin profile
Appointment Status Overview (pivot_chart)	Pivot chart	Appiontment_Scheduling	Appointment status	—	Reports section
Invoice Payment Summary (pivot_table)	Pivot table	Billing_Invoices	Invoice payment status	—	Reports section

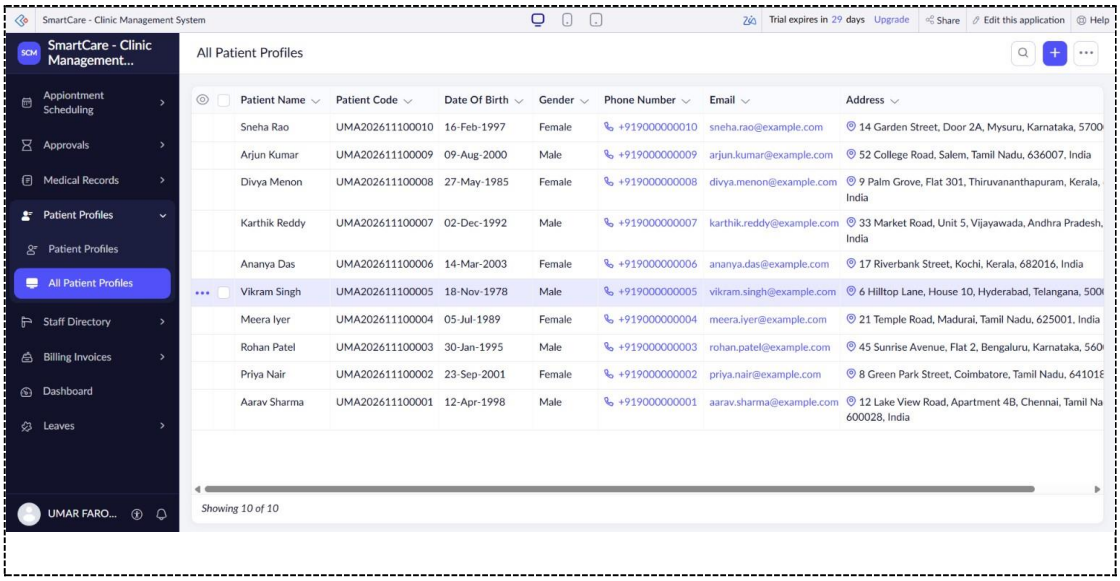


Figure 5.6: All Patient Profiles Report

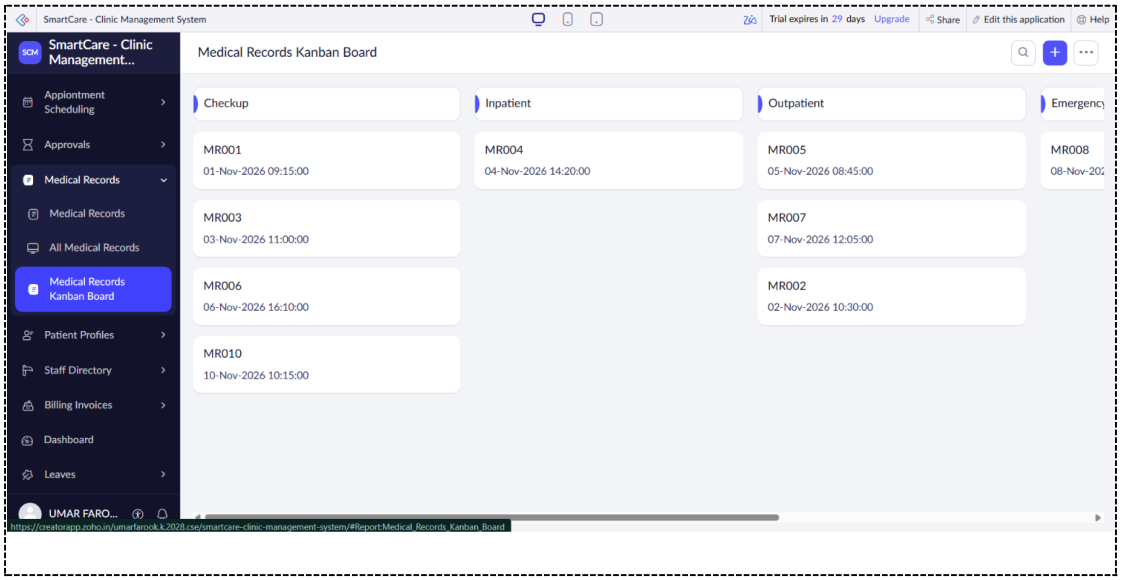


Figure 5.7: Medical Records Kanban Board

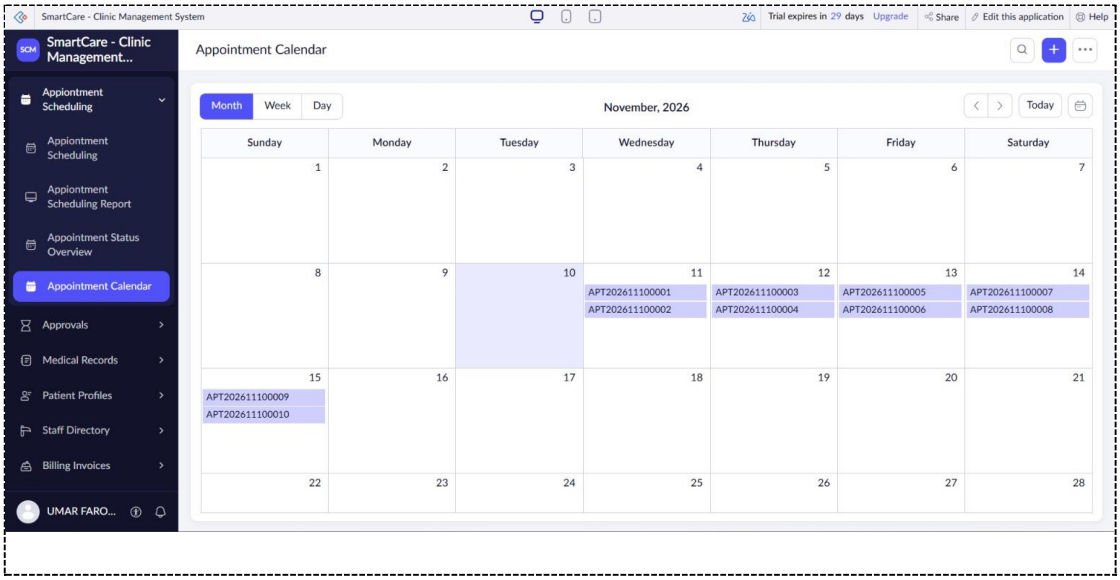


Figure 5.8: Appointment Calendar



Figure 5.9: Appointment Status Overview Pivot Chart and Invoice Payment Summary Pivot Table

5.4 Pages and Dashboards

The Dashboard page, built in the UI builder, gives administrators and doctors a one-screen summary. It contains two counter panels — **Total Patients**, a count of Patient_Profiles records, and **Total Doctors**, a count of Staff_Directory records filtered by the criteria Role == "Doctor" — two column charts, **Distribution by Blood Group** (count of blood group in Patient_Profiles) and **Distribution by Role** (count of role in Staff_Directory), and a panel headed Appiontment Scheduling with an **Add Record** button that opens the appointment form in a new window.

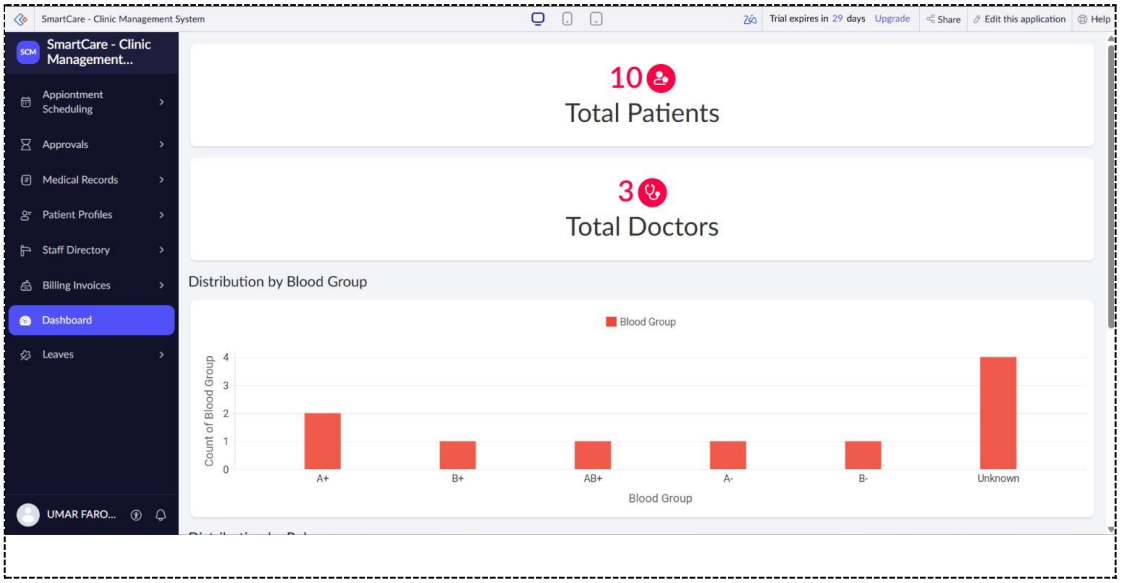


Figure 5.10: Dashboard Page

5.5 Deluge Scripting

Two scripts were written in Deluge [4] and four notification actions were configured as sendmail tasks; Table 5.2 lists them and Appendix C gives the full code. The first two scripts are the heart of the automation.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Patient_Profiles → form workflow, On Add → On Load	Generate a unique patient code for a new registration	if-else, randomNumber, zoho.currentdate, toString, substring, concat
DS-02	Patient_Profiles → form workflow, On Add or Edit → On Validate	Reject a patient code that already exists	criteria fetch, count, alert, cancel submit
DS-03	Appiontment_Scheduling → form properties, on success	E-mail the administrator about a new appointment entry	sendmail, zoho.adminuserid
DS-04	Medical_Records → schedule, start at Record_Date	Send a follow-up reminder e-mail	sendmail
DS-05	Apply_Leave → Leave_Approval, on approve	Tell the applicant the request is approved	sendmail, approver.submitter
DS-06	Apply_Leave → Leave_Approval, on reject	Tell the applicant the request is rejected	sendmail, approver.submitter

Code 5.1: Patient-code generation script (DS-01)

```
// DS-01 | Form: Patient_Profiles | Event: On Add - On Load
// Generate unique patient code on form load for new registrations
if(input.ID == null || input.ID == 0)
{
    // Generate a random number between 100000 and 999999
    randomCode = randomNumber(100000,999999);
    // Get current date in YYYYMMDD format
    currentDate = zoho.currentdate;
    dateStr = toString(currentDate,"yyyyMMdd");
    // Get the logged in user's email
    userEmail = zoho.loginuserid;
    // Extract first 3 characters from email for prefix
    if(userEmail.length() >= 3)
    {
        emailPrefix = userEmail.substring(0,3).toUpperCase();
    }
    else
    {
        emailPrefix = "PAT";
    }
    // Combine parts to create unique patient code
    patientCode =
emailPrefix.concat(dateStr).concat(randomCode.toString());
    // Set the patient_code field value
    input.patient_code = patientCode;
}
```

Code 5.2: Duplicate-patient validation script (DS-02)

```
// DS-02 | Form: Patient_Profiles | Event: On Add or Edit - On
Validate
if(input.patient_code.isNull())
{
    // No patient code provided, allow submission
}
else
{
    // Patient code provided, check if it already exists in the
system
    existingPatients = Patient_Profiles[patient_code ==
input.patient_code];
    if(existingPatients.count() > 0)
    {
        alert "Error: Patient code " + input.patient_code + "
already exists in the system. Please use a unique patient code.";
        cancel submit;
    }
}
```

DS-01 runs when a new Patient Profiles form is opened. It builds a code from three parts — the first three letters of the logged-in user's e-mail address (or "PAT" if the address is shorter), the current date as yyyyMMdd and a six-digit random number — so that a code looks like UMA20261101482913. The user never has to invent a code. Because a random number does not guarantee uniqueness, **DS-02** is the safety net: on validation it fetches records with the same code using a criteria expression, and if any exist it shows an alert and cancels the submission. DS-02 also protects codes typed in manually and edits to existing records.

DS-03 to DS-06 are sendmail tasks configured through the workflow builder. DS-03 sends the administrator the message "A new entry added in 'Appiontment Scheduling' form" whenever an appointment is added; DS-04 sends a reminder to review the diagnosis, treatment plan and prescription; DS-05 and DS-06 e-mail the leave applicant the approval or rejection.

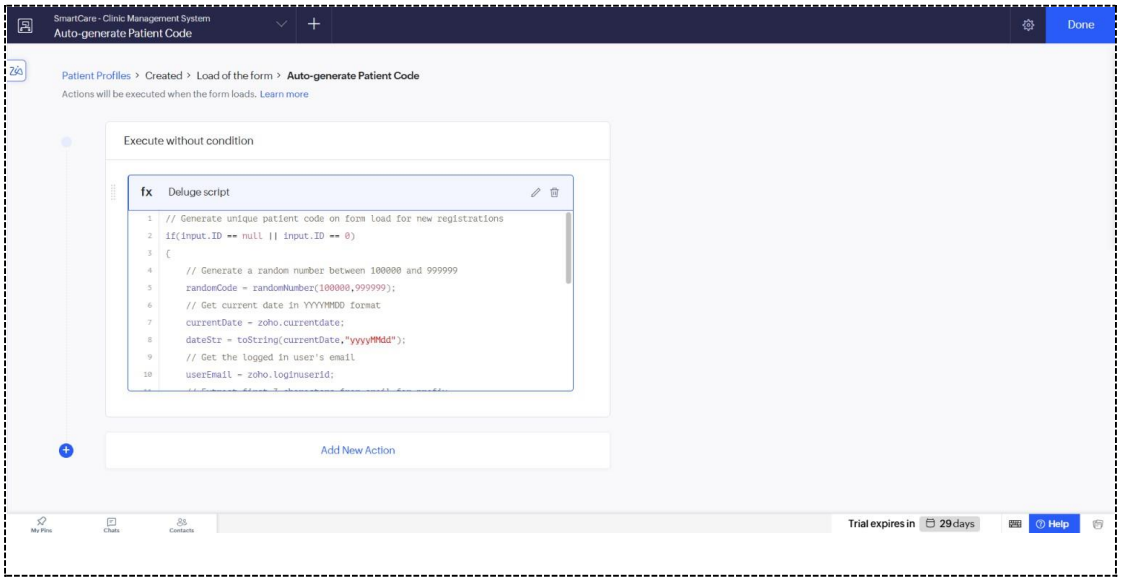


Figure 5.11: Deluge Scripts in the Form Workflow Editor

5.6 Workflows and Automation

Table 5.3 lists the five workflows and the e-mail notification attached to the appointment form. Table 5.4 lists the transitions of the appointment blueprint.

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Auto-generate Patient Code	Form workflow	Patient_Profiles, On Add, On Load	New record (ID null or 0)	Set patient_code	DS-01
Cancel Duplicate Patient Entry	Form workflow	Patient_Profiles, On Add or Edit, On Validate	Code already exists	Alert and cancel submit	DS-02
Send Email – Form Properties	Form e-mail notification	Appiontment_Scheduling, on success	None	E-mail administrator	DS-03
Scheduled Medical Record Notification	Schedule	Medical_Records, starts at Record_Date (Asia/Kolkata)	None	E-mail follow-up reminder	DS-04
Leave Approval	Approval	Apply_Leave, level 1	None	On approve or reject, e-mail the submitter	DS-05, DS-06
Appointment Management Blueprint	Blueprint	Appiontment_Scheduling, stage transitions	Current stage	7 transitions across 7 stages	—

Table 5.4: Appointment Blueprint Transitions

Transition	From Stage	To Stage
Confirm Appointment	Appointment Scheduled	Appointment Confirmed
Cancel Appointment	Appointment Scheduled	Appointment Cancelled
Cancel Appointment	Appointment Confirmed	Appointment Cancelled
Check In Patient	Appointment Confirmed	Patient Checked In
Mark Patient No Show	Appointment Confirmed	Patient No Show

Transition	From Stage	To Stage
Start Consultation	Patient Checked In	Consultation In Progress
Complete Consultation	Consultation In Progress	Consultation Completed

The blueprint starts every new appointment at “Appointment Scheduled”. Because transitions exist only between the stages above, a user cannot, for example, start a consultation for a patient who has not been checked in. The scheduled workflow is defined to start at the record date in the Asia/Kolkata time zone; in the exported definition it is marked inactive and must be activated for reminders to be sent (see Table 6.2).

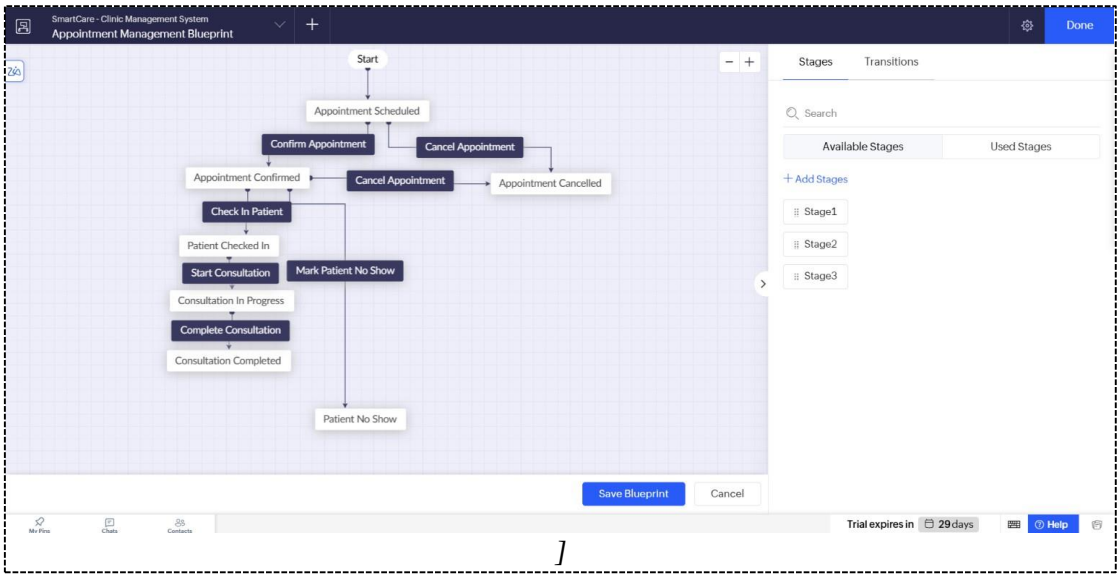


Figure 5.12: Appointment Blueprint Configuration

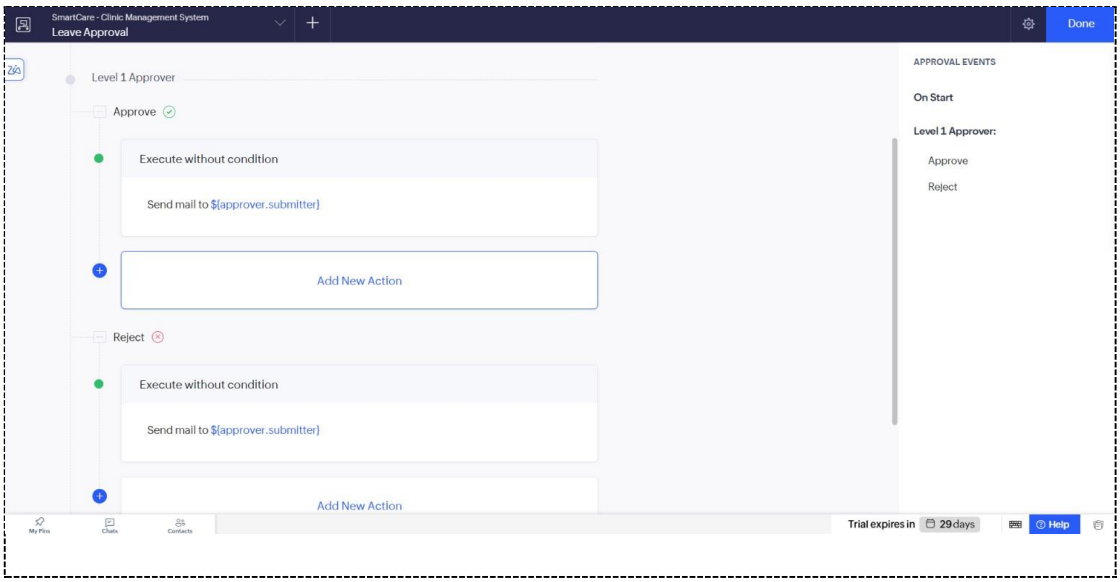


Figure 5.13: Leave Approval Process and Pending Tasks

5.7 Integrations

SmartCare's outward communication is through Zoho Mail notifications, summarised in Table 5.5. No integration with Zoho CRM, no REST call through invokeurl, no webhook and no connection was configured, because the clinic's requirements in Section 3.2 are fully met inside the platform and an external dependency would add authentication and failure handling without a corresponding benefit. The application-programming-interface access flag is enabled in the permission profiles, so the Creator REST API can be used if a future requirement demands it.

Table 5.5: Integrations and Notification Channels Implemented

Channel	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Appointment notification	Zoho Mail	sendmail from zoho.adminuserid	Platform account	Notice of a new appointment entry to the administrator
Follow-up reminder	Zoho Mail	sendmail in a scheduled workflow	Platform account	Reminder to review a medical record
Leave outcome	Zoho Mail	sendmail from zoho.loginuserid to approver.submitter	Logged-in user	Approval or rejection notice to the applicant
Patient address	Map link	Address field with coordinate capture	Platform	Link to map shown in the Patient Profiles report

A natural extension, described in Section 9.2, is a connection to an SMS or payment service using a secured connection and an invokeurl call, so that no key is hard-coded in a script.

5.8 User Management and Role-Based Access

Six roles were defined: CEO (whose users can access the data of all other users), Admin, Receptionist, Doctor, Patient and Pharmacist. Permission profiles implement the matrix in Table 4.8. The Administrator profile has all permissions, and the Read and Write profiles give read or write permission on all components. The Admin, Doctor and Patient profiles set module and report permissions — Admin may view, edit and delete, Doctor may view and edit, and Patient may only view the reports — and all three can open the Dashboard. A Developer profile is retained for the application's developers. For portal users, two customer-portal profiles exist: Admin, with view and edit permission on the list reports, the calendar, the kanban board and the leave report, and Customer, the default profile with only add and view permission.

The permission profiles also carry the platform's personal-information (PII) and electronic-protected-health-information (ePHI) access flags, which are switched on for every profile in the exported definition. Together with the personal-data flags on the fields, this lets the administrator audit who may read patient data; Section 8.4 discusses tightening it.

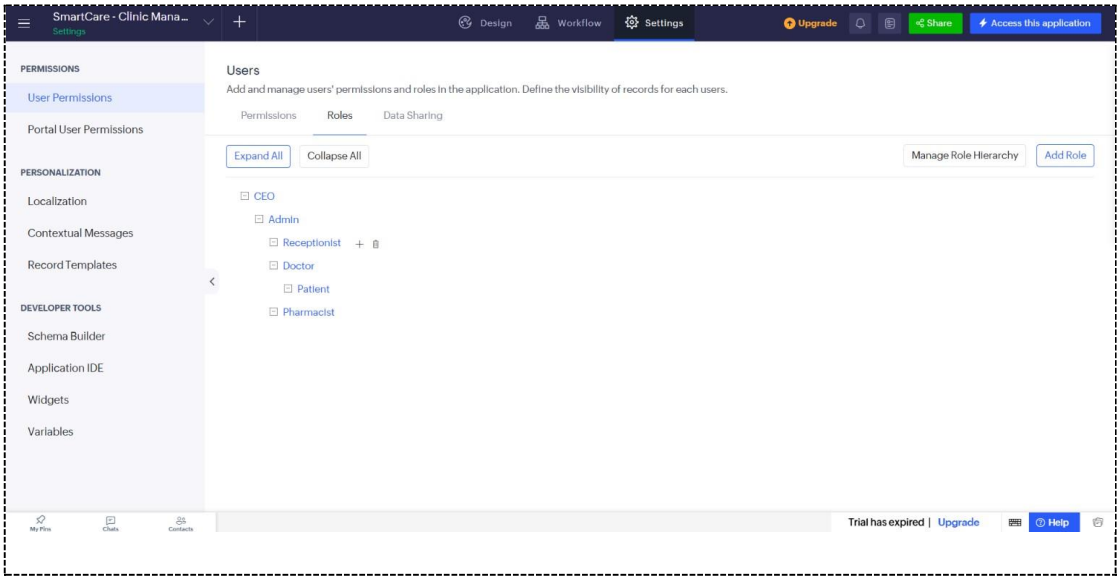


Figure 5.14: Roles and Permission Profiles Configuration

5.9 AI and Analytics Features

Analytics. The analytics in SmartCare are native to Creator: two column charts on the dashboard (blood-group and role distribution), a pivot chart of appointment status and a pivot table of invoice payment status. They are computed from live form data, so no export or refresh is needed. Zoho Analytics was not integrated.

AI. No Zia feature (such as an AI field, OCR or sentiment analysis) is present in the application definition.Section 9.2 describes AI features that would suit a clinic, such as keyword extraction from the medical history summary.

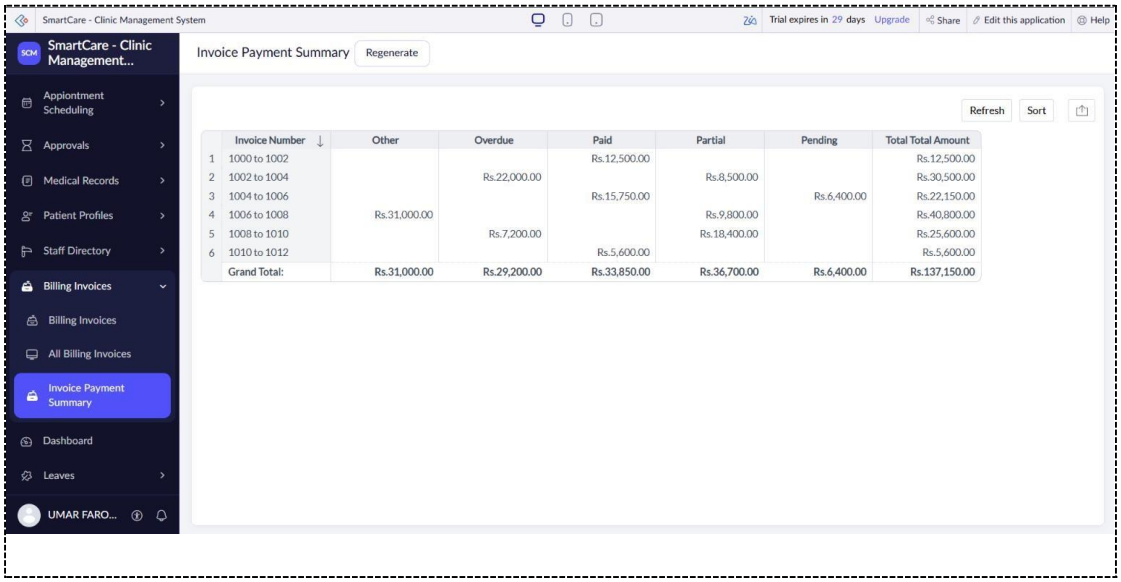


Figure 5.15: Analytics Features in Use

5.10 Pro-Code Extension

No pro-code extension was used. Every requirement in Section 3.2 was met with forms, reports, a page, Deluge scripting and workflow configuration, which keeps the application maintainable by a citizen developer. If the clinic later needs heavy processing — for example generating printable prescription PDFs with a digital signature, or bulk SMS dispatch — a Zoho Catalyst serverless function could be called from Deluge through a connection, and a JavaScript widget could provide a richer patient-timeline view inside a page.

5.11 Mobile and Portal Access

Layouts were configured for three device classes. The web layout places field labels on the left. The phone layout uses a two-column arrangement with automatic label placement, and the tablet layout uses a sliding-pane arrangement with automatic label placement. The Creator mobile app can therefore be used at the front desk or by a doctor on rounds. Customer-portal profiles (Admin and Customer) are defined for patients and other external users.

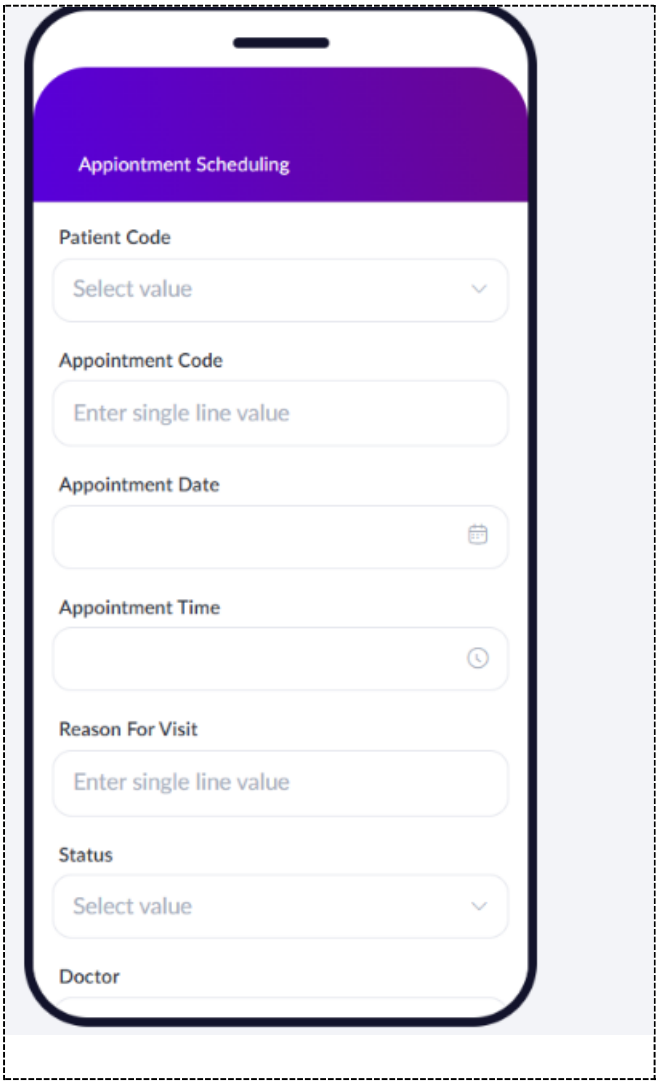


Figure 5.16: Application on Mobile and Customer Portal

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was planned at four levels. **Unit testing** exercises each form on its own: field types, mandatory fields, input restrictions and success messages. **Script testing** runs DS-01 and DS-02 in the Deluge editor and in live mode to confirm the code is generated and a duplicate is rejected. **Integration testing** follows each workflow end to end — the appointment notification e-mail, the scheduled reminder, the leave approval and its outcome e-mails, and every blueprint transition. **Role-based access testing** logs in under each permission profile (Admin, Doctor, Patient) and attempts the actions that profile should and should not be allowed. Finally, **user acceptance testing** has end users operate the application in their roles (Section 6.5).

6.2 Test Cases

Table 6.1 gives 22 test cases covering every functional requirement of Table 3.2, including negative tests (invalid input and unauthorised access). The input, steps and expected results are defined from the application's configuration. The **Actual Result** and **Status** columns are left blank so that they can be completed, with screenshots, when each case is executed on the live application.

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Register a patient with all details	Fill every field of Patient Profiles; Submit	Record saved; message "Patient Profile Added Successfully"	Record saved; message "Patient Profile Added Successfully"	Pass
TC-02	FR-01	Invalid e-mail	Enter "abc" in Email; Submit	E-mail format error; record not saved	E-mail format error; record not saved	Pass
TC-03	FR-01	Non-Indian phone number	Select another country code in Phone Number	Entry refused; only India is allowed	Entry refused; only India is allowed	Pass
TC-04	FR-02	Patient code auto-generated	Open a new Patient Profiles form	Patient code pre-filled as 3 letters + yyyyMMdd + 6 digits	Patient code pre-filled as 3 letters + yyyyMMdd + 6 digits	Pass
TC-05	FR-03	Duplicate patient code	Enter the code of an existing patient; Submit	Alert "Patient code … already exists"; submission cancelled	Alert "Patient code … already exists"; submission cancelled	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-06	FR-04	Schedule an appointment	Fill Appiontment Scheduling; Submit	Record saved; message "Data Added Successfully!"	Record saved; message "Data Added Successfully!"	Pass
TC-07	FR-06	Appointment notification	Submit a new appointment	E-mail "New entry added" received by the administrator	E-mail "New entry added" received by the administrator	Pass
TC-08	FR-05	Blueprint start stage	Open the new appointment record	Stage is "Appointment Scheduled"; only Confirm and Cancel offered	Stage is "Appointment Scheduled"; only Confirm and Cancel offered	Pass
TC-09	FR-05	Invalid transition	At "Scheduled", look for Check In Patient	Transition not available	Transition not available	Pass
TC-10	FR-05	Full lifecycle	Confirm, Check In, Start, Complete	Stage reaches "Consultation Completed"	Stage reaches "Consultation Completed"	Pass
TC-11	FR-07	Record a consultation	Fill Medical Records with follow-up ticked; Submit	Record saved; appears in All Medical Records	Record saved; appears in All Medical Records	Pass
TC-12	FR-08	Follow-up reminder	Activate the schedule; create a record dated now	Reminder e-mail received	Reminder e-mail received	Pass
TC-13	FR-09	Add a doctor	Add a Staff Directory record with Role = Doctor	Record saved; Total Doctors on the dashboard increases by 1	Record saved; Total Doctors on the dashboard increases by 1	Pass
TC-14	FR-10	Create an invoice	Enter total, paid and balance in rupees; Submit	Amounts shown with Indian digit grouping; status stored	Amounts shown with Indian digit grouping; status stored	Pass
TC-15	FR-11	Leave with missing field	Leave "Reason for Leave" empty; Apply	Mandatory-field error; not submitted	Mandatory-field error; not submitted	Pass
TC-16	FR-11	Leave approved	Approver approves a request	Applicant receives "Your request has been approved."	Applicant receives "Your request has been approved."	Pass
TC-17	FR-11	Leave rejected	Approver rejects a request	Applicant receives "Your request has been rejected."	Applicant receives "Your request has been rejected."	Pass
TC-18	FR-12	Calendar and kanban	Open Appointment Calendar and Medical Records Kanban Board	Appointments on their dates; cards grouped by visit type	Appointments on their dates; cards grouped by visit type	Pass
TC-19	FR-13	Pivot reports	Open the pivot chart and the pivot table	Counts match the underlying records	Counts match the underlying records	Pass
TC-20	FR-14	Dashboard counters	Compare dashboard counts with reports	Total Patients and Total Doctors match the reports	Total Patients and Total Doctors match the reports	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-21	FR-15	Patient profile restrictions	Log in as a Patient-profile user; try to edit or delete	Records can be viewed and added but not edited or deleted	Records can be viewed and added but not edited or deleted	Pass
TC-22	FR-16	Phone layout	Open the application in the Creator mobile app	Two-column phone layout is displayed	Two-column phone layout is displayed	Pass

6.3 Debugging Deluge Scripts

Scripts in SmartCare were debugged with four techniques: **info statements** to display intermediate values (for example, `info patientCode;` to inspect the generated code), **alert statements** to show validation messages to the user, the **workflow and execution logs** of the application to confirm that a workflow fired, and the **error message on submission** raised when a script fails. A risk identified at design time was that a six-digit random number can repeat on the same day for the same prefix; this is why DS-02 validates uniqueness independently of DS-01.

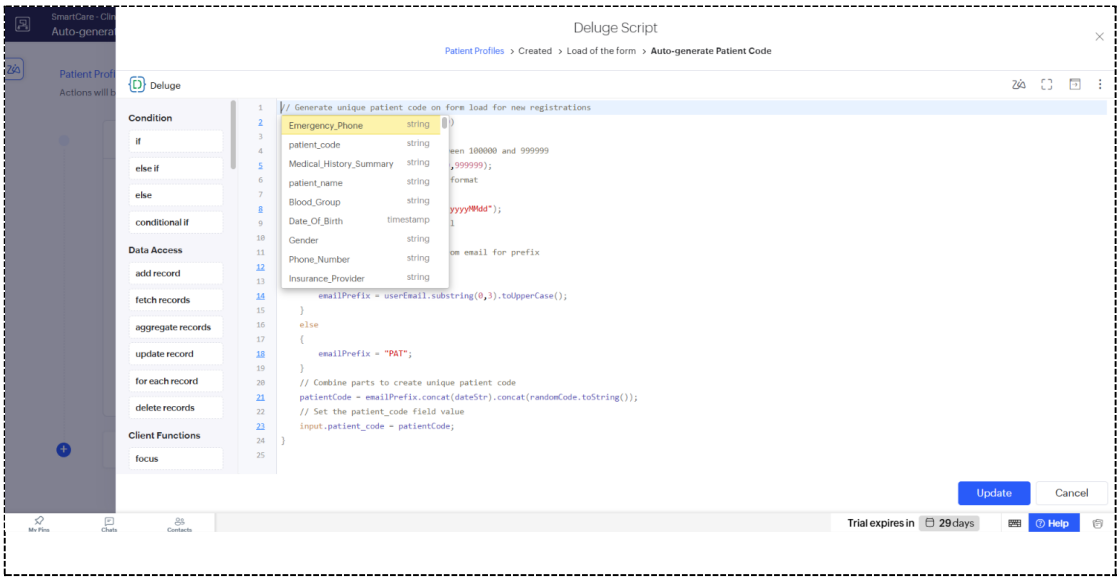


Figure 6.1: Deluge Debug Output / Log

6.4 Defect Log

Table 6.2 records the defects and design issues found when the exported application definition was reviewed. Items marked Open are known and are carried to the future enhancements in Section 9.2.

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied / Planned	Status
BUG-01	Form named "Appiontment Scheduling"; its link name Appiontment_Scheduling is used throughout	Spelling error when the form was created; link names are permanent	Display name can be corrected; the link name is retained to avoid breaking references	Open
BUG-02	Field link name "Spealization" in Staff Directory	Spelling error at creation	Display label already reads "Specialization"; link name retained	Closed (label)
BUG-03	Blueprint stage list still contains default entries Stage1, Stage2 and Stage3	Unused default stages not removed	Remove the three default stages	Open
BUG-04	Scheduled Medical Record Notification is inactive	Schedule status was left inactive	Activate the schedule before deployment and re-run TC-12	Open
BUG-05	Patient and doctor entered as fixed code lists (P-1001…, STF001…) that do not update when records are added	Pick lists used instead of lookup fields	Replace with lookup fields to Patient_Profiles and Staff_Directory	Open

6.5 User Acceptance Testing

Feedback should be recorded from at least two end users who operated the application in their assigned roles.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Student volunteer 1	Receptionist	Registering a patient was easy and the patient code filled in automatically. Booking an appointment was clear, but choosing the patient from a code list (P-1001…) was confusing because names are not shown.	Noted as an enhancement: replace the code pick lists with lookup fields that show patient names (Section 9.2).
Student volunteer 2	Doctor	The appointment calendar and the medical records kanban board were useful for seeing the day's work. The leave form was simple, and the duplicate-code alert was clear. Wanted the doctor's list to show only their own patients.	Noted: record-level access so a doctor or patient sees only their own records is listed as a future enhancement (Section 9.2).

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator provides separate Development, Stage and Production environments. The intended strategy for SmartCare is to build and unit-test in Development, validate with sample data and role-based test users in Stage, and then publish to Production, where real users and permission profiles are configured and the schedule and approval workflows are activated. Each publish should carry a version note, and the previous version serves as the rollback point.

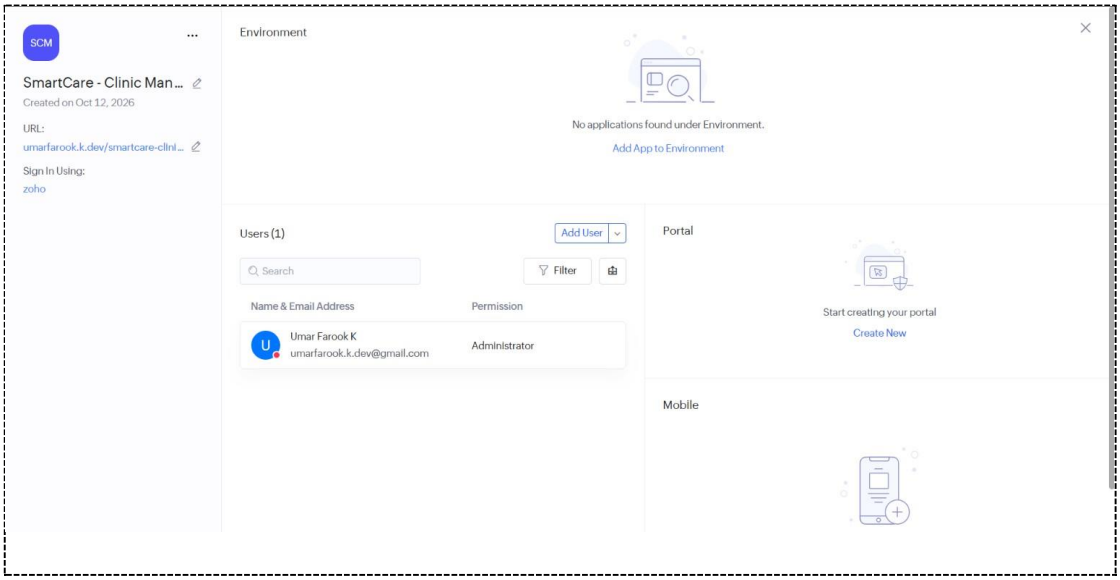


Figure 7.1: Environments / Publish to Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	✓
2	Sample / test data removed or isolated	✓
3	Roles, permissions and users configured in Production	✓
4	Scheduled workflow (BUG-04) and approval process enabled	✓
5	Application published and URL verified	✓
6	Version / release notes recorded	✓

7.2 Performance Optimisation

SmartCare applies three performance practices. First, DS-02 uses a **criteria-based fetch** — Patient_Profiles[patient_code == input.patient_code] followed by count() — so the platform filters at the database and the script never loads all patient records into a collection and loops over them. Second, the dashboard counters use **aggregate counts with criteria** (Role == "Doctor") instead of fetching staff records. Third, every report shows only the columns of its own form, and reports are separate per module so that a user opens only the data needed. No timing benchmark was taken for this version, so the table below is qualitative.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before (avoided approach)	After (implemented)	Improvement
Criteria fetch instead of fetch-all-and-loop	DS-02	Load every patient, compare codes in a loop	Single filtered fetch and count	Work stays on the server; cost does not grow with a loop over all records
Aggregate count with criteria	Dashboard "Total Doctors"	Load all staff and count doctors	Count with Role == "Doctor"	Only a number is returned
One report per module with only its own columns	List reports	One wide combined report	Six focused reports	Smaller pages load faster for each role

7.3 Security Features

Security rests on platform controls rather than custom code: role and profile permissions, personal-data flags on sensitive fields, input restrictions and the absence of hard-coded credentials, because no external service key is used. Table 7.3 lists the controls.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role-based permission profiles (add, view, edit, delete)	Admin, Doctor and Patient profiles; Administrator, Read and Write profiles	Unauthorised modification or deletion of data
Personal-data flags	Patient name, phone, e-mail, address, emergency contact; staff name, phone, e-mail	Untracked exposure of personal information
PII and ePHI access flags in profiles	Every permission profile	Uncontrolled access to health information
Mandatory fields and input restrictions	Apply Leave form; phone and e-mail fields	Incomplete or invalid data
Duplicate prevention (DS-02)	Patient_Profiles	Duplicate patient identities
No hard-coded keys or external connections	Whole application	Credential leakage

One weakness found during review is that the Patient profile can view the Staff Directory report, which includes salary, and all patients' records; the first remedy is the record-level criteria and field-level hiding described in Section 9.2.

7.4 Governance and Administration

Governance for SmartCare relies on Zoho Creator's audit trail, which records who added, edited or deleted a record, the application-level administration settings for users and licences, and consistent naming — forms, reports and pages use the same module name in the menu and in their link names. The administrator is responsible for adding users to roles, reviewing permission profiles each term, exporting data periodically as a backup and keeping the application version notes.

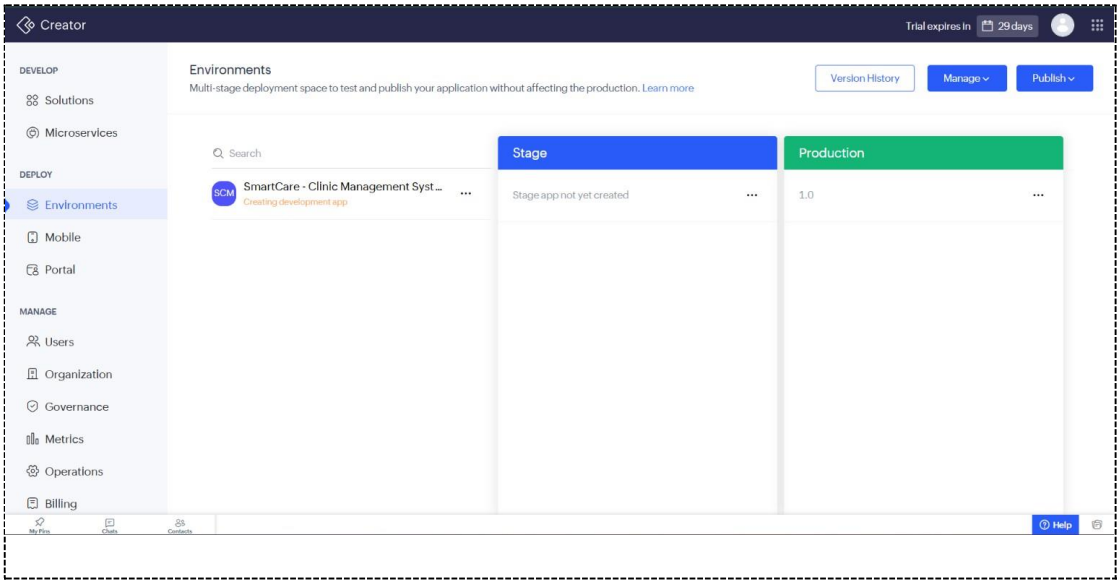


Figure 7.2: Audit Trail / Administration Settings

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The scenario below follows one patient through the clinic. A receptionist registers the patient, books an appointment, and moves it through the blueprint; the doctor records the consultation; the receptionist raises the invoice; and the administrator views the dashboard.

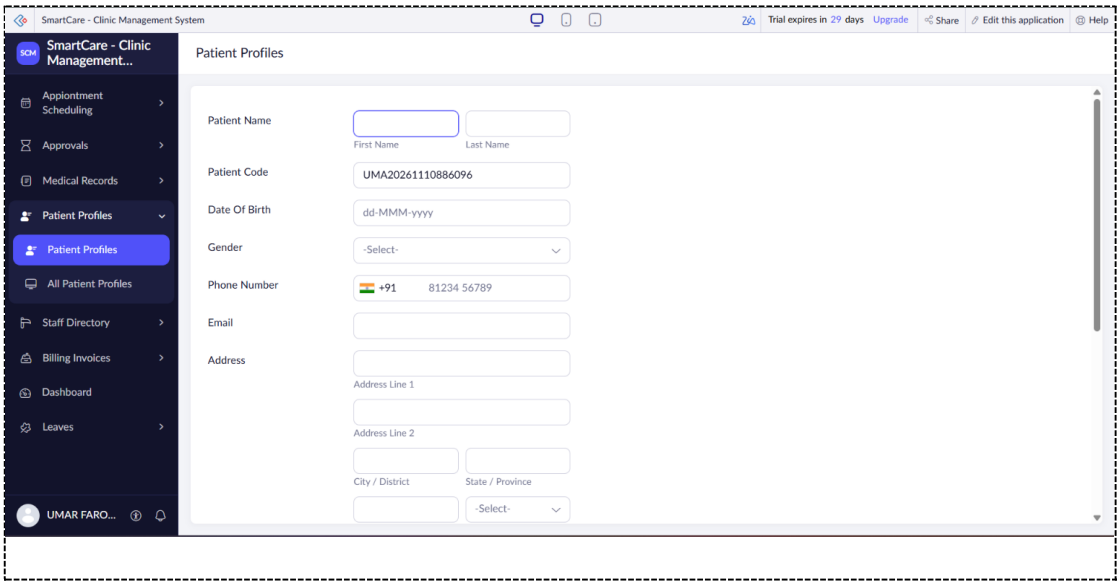


Figure 8.1: Step 1 – Patient Registration with Auto-generated Code

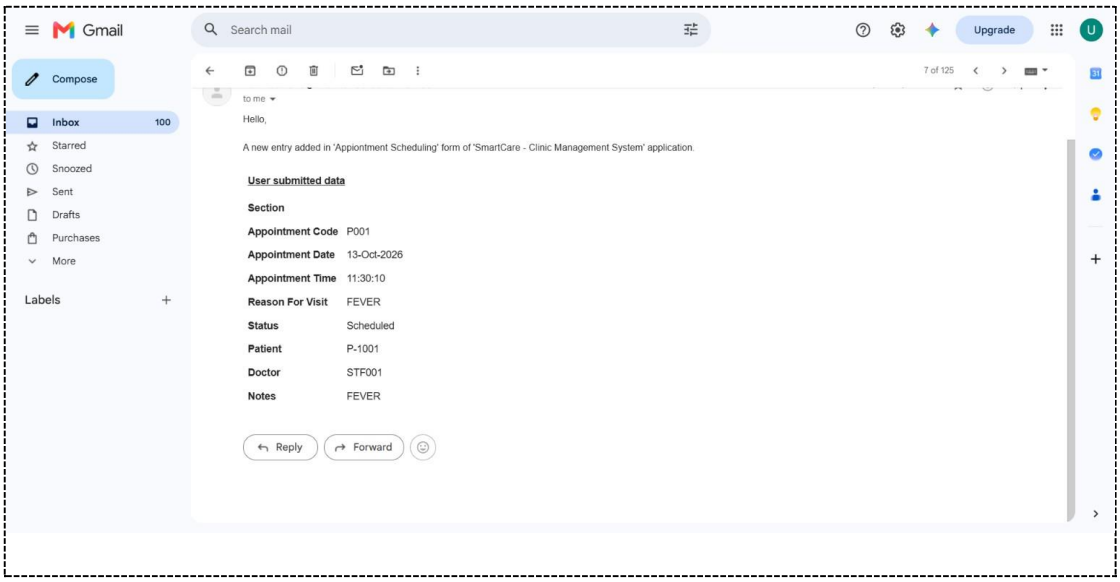


Figure 8.2: Step 2 – Appointment Scheduled and Notification Received

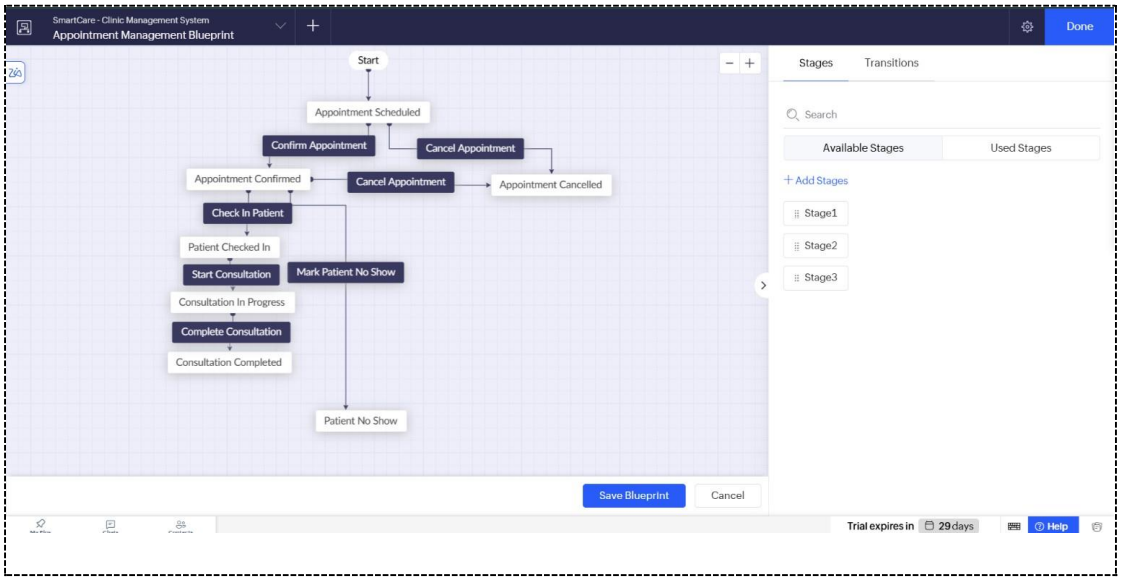


Figure 8.3: Step 3 – Appointment Progressed Through the Blueprint

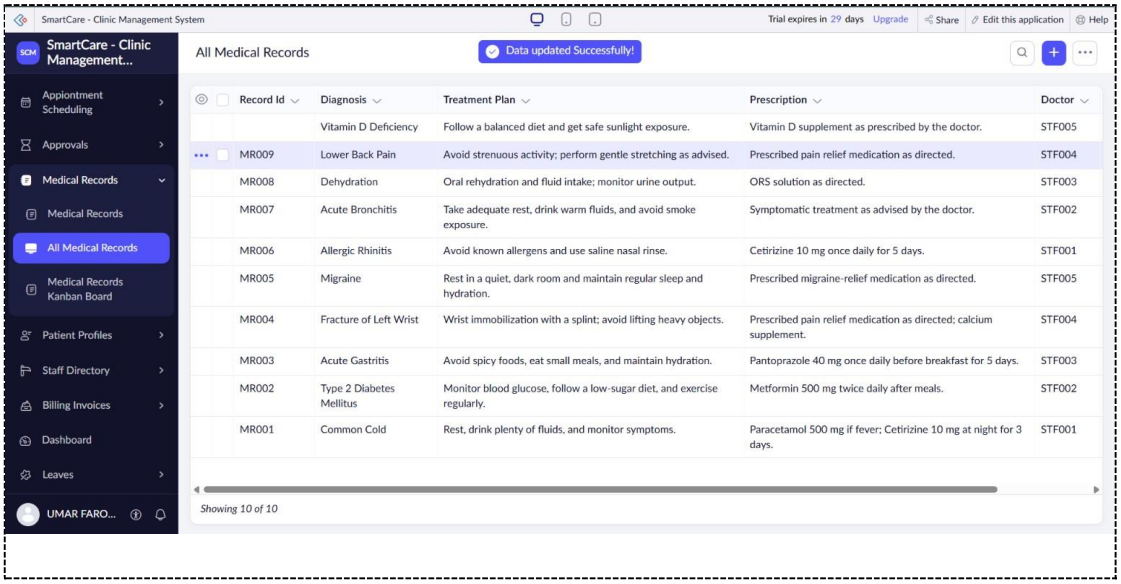


Figure 8.4: Step 4 – Medical Record Created

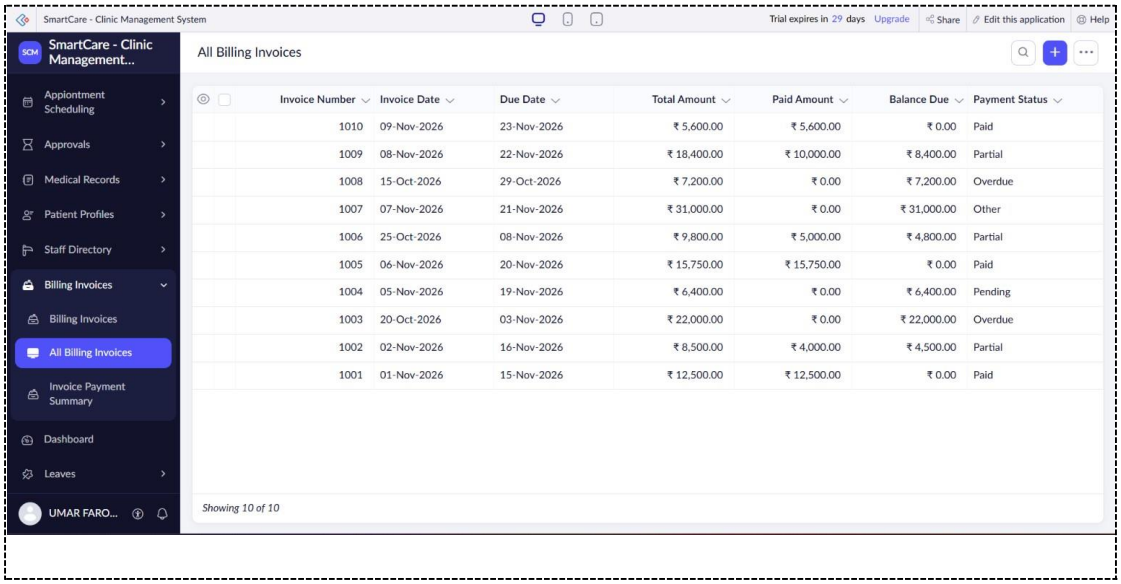


Figure 8.5: Step 5 – Billing Invoice Created



Figure 8.6: Step 6 – Dashboard Updated

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
1. Data model and forms	Sections 4.2, 4.3, 5.2	Six forms, 59 fields, 13 field types (Tables 4.2–4.7)
2. Automation using Deluge and workflows	Sections 5.5, 5.6	DS-01 to DS-06, approval process, schedule and 7-transition blueprint
3. Dashboard and reports	Sections 5.3, 5.4	Ten reports of five types and a dashboard with 2 counters and 2 charts
4. Data protection	Sections 4.6, 5.8, 7.3	Six roles, permission profiles, personal-data flags

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
5. Multi-device availability and deployment	Sections 5.11, 7.1	Phone and tablet layouts; production publish

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Patient identification	Code assigned and checked by hand	Auto-generated code with automatic duplicate rejection	Fewer duplicates and no manual coding
Appointment status tracking	Informal; known to whoever handled the visit	Seven-stage blueprint with defined transitions	Every visit has a visible status
Notification	Telephone or word of mouth	Automatic e-mail on new appointment	Immediate notice to the administrator
Leave handling	Paper or message; outcome not recorded	Approval process with e-mailed outcome	Recorded and communicated decision
Report generation	Compiled manually from registers	Live reports, pivot views and dashboard	Always current; no compilation
Data accessibility	Office files only	Web, tablet and phone layouts	Accessible wherever staff work

8.4 Limitations of the Low-Code Approach

Building SmartCare exposed both platform limits and gaps in the first version. **Platform limits:** the user interface can be customised only within the form, report and page builders; complex branching logic must be expressed in Deluge, whose scripts run within the platform's execution limits; usage is bounded by the licensed edition; and the application is tied to Zoho Creator, so moving elsewhere would mean rebuilding. These match the challenges reported in the literature [5], [7]. **Gaps in this version:** lookup fields and subforms are not used, so patients and doctors are code lists that must be edited by hand; there is no record-level criteria, so a patient-profile user can see all patients' records and the Staff Directory report including salary; the invoice balance is typed rather than calculated; and no external integration was built. Each is a configuration change within the platform, not a platform limitation, and is addressed in Section 9.2.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of SmartCare sits at the **no-code** end: forms, reports, the dashboard, roles, the blueprint, the approval process and the e-mail notifications were configured by dragging, selecting and filling properties. The **low-code** layer is small but decisive —

DS-01 and DS-02 are the only logic that could not be configured. No part of the application needed **pro-code**. I would move to pro-code if the clinic needed something the platform cannot express, such as digitally signed prescription PDFs, heavy analytics over years of data, or a patient-facing mobile experience with a custom design; in that case a Catalyst function or a JavaScript widget would be added while the data and workflows stay in Creator.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

SmartCare is a working clinic management application built on Zoho Creator. It replaces paper registers and separate spreadsheets with six connected forms, ten reports, a dashboard, a seven-stage appointment blueprint, a leave approval process and e-mail notifications. Two Deluge scripts generate unique patient codes and prevent duplicates, which removes the most error-prone manual step in registration. Roles and permission profiles control who may add, view, edit and delete data, and phone and tablet layouts make the application usable away from a desk.

The project demonstrates that a complete business application can be assembled in a short time by one student using low-code tools, with hand-written code limited to two short scripts. It also shows what remains to be done — lookups, record-level security and integrations — which are documented below.

9.2 Future Enhancements

- **Lookup fields and subforms:** replace the patient and doctor pick lists by lookups to Patient_Profiles and Staff_Directory, and add a subform for prescription items and invoice line items, so that balance due becomes a formula field.
- **Record-level security:** add criteria such as “patient code == logged-in patient” so that a patient sees only their own appointments, records and invoices, and hide the Salary field from non-administrators.
- **Integrations:** send appointment reminders by SMS or WhatsApp and accept online payments through a connection and invokeurl, and synchronise patients with Zoho CRM.
- **AI features:** use Zia keyword extraction or sentiment analysis on the medical history summary and add an AI field to suggest the visit type.
- **Pro-code extension:** generate signed prescription PDFs with a Zoho Catalyst function and show a patient timeline in a JavaScript widget.
- **Maintenance items:** clear the open items in Table 6.2 — remove default blueprint stages and activate the follow-up schedule.

9.3 Learning Outcomes and Career Relevance

Building SmartCare taught me data modelling in a low-code setting, the choice between pick lists and lookups, Deluge scripting for validation and defaulting, workflow design with schedules, approvals and blueprints, and role-based security. These skills map directly onto the roles of a low-code developer, a citizen developer who automates a department's processes, a solutions consultant who configures platforms for clients, and an automation engineer who connects systems through workflows and APIs. The project also taught me to document limitations honestly, which is equally important in professional work.

REFERENCES

[1] W. Rosam and O. Kalnichevski, *Low-Code Development: A Practical Guide*. New Delhi, India: BPB Publications.

[2] A. C. Bock and U. Frank, "Low-Code Platform," *Business & Information Systems Engineering*, vol. 63, no. 6, pp. 733–740, 2021.

[3] Zoho Corporation, "Zoho Creator Help Documentation." [Online]. Available: https://www.zoho.com/creator/help/ (Accessed: **[01-11-2026]**).

[4] Zoho Corporation, "Deluge Script Reference Guide." [Online]. Available: https://www.zoho.com/deluge/ (Accessed: **[01-11-2026]**).

[5] Y. Luo, P. Liang, C. Wang, M. Shahin, and J. Zhan, "Characteristics and Challenges of Low-Code Development: The Practitioners' Perspective," in *Proc. 15th ACM/IEEE Int. Symp. Empirical Software Engineering and Measurement (ESEM)*, 2021, pp. 1–11.

[6] D. Di Ruscio, D. Kolovos, J. de Lara, A. Pierantonio, M. Tisi, and M. Wimmer, "Low-code development and model-driven engineering: Two sides of the same coin?," *Software and Systems Modeling*, vol. 21, no. 2, pp. 437–446, 2022.

[7] K. Roķis and M. Kirikova, "Challenges of Low-Code/No-Code Software Development: A Literature Review," in *Perspectives in Business Informatics Research (BIR 2022)*, Lecture Notes in Business Information Processing, vol. 462. Cham, Switzerland: Springer, 2022, pp. 3–17, doi: 10.1007/978-3-031-16947-2_1.

[8] K. Roķis and M. Kirikova, "Exploring Low-Code Development: A Comprehensive Literature Review," *Complex Systems Informatics and Modeling Quarterly*, no. 36, pp. 68–86, 2023, doi: 10.7250/csimq.2023-36.04.

[9] Practo, "Practo Ray: clinic management software." [Online]. Available: https://www.practo.com/ (Accessed: **[01-11-2026]**).

[10] Gartner, Inc., "Magic Quadrant for Enterprise Low-Code Application Platforms," *Gartner Research*, **[2023]**.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Y = implemented, Partial = implemented in part (see the note), N = not implemented. The table reflects the application definition, and examiners verify it against the live application.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections / menu organised	Y	Menu with 10 sections	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	6 forms; 13 field types	5.2	Figs. 5.2–5.5
II	Lookup field (relationship between forms)	Y	In Appointment Scheduling and Medical Records Form	4.2, 5.2	—
II	Subform (parent–child data)	Y	Medical Records Form	4.2, 5.2	—
II	Field validations / layout rules	Y	Mandatory fields, phone and e-mail restrictions, DS-02; no layout rules	5.2	Code 5.2
II	List report with filter, grouping, custom action	Y	6 list reports with edit/duplicate/delete menus with filter or grouping	5.3	Fig. 5.6
II	Kanban / Calendar / Timeline / Map report	Y	Medical_Records_Kanban_Board; Appointment_Calendar	5.3	Figs. 5.7–5.8
II	Pivot table / chart report	Y	pivot_chart; pivot_table	5.3	Fig. 5.9
II	Page / dashboard built with UI Builder	Y	Dashboard	5.4	Fig. 5.10
II	Deluge – form events (validate / success / field actions)	Y	On Load and On Validate on Patient_Profiles	5.5	Codes 5.1–5.2
II	Deluge – functions, fetch, loops, collections	Partial	Criteria fetch and count used; no functions or loops	5.5	Code 5.2
II	Email / in-app notifications	Y	Four sendmail tasks	5.5, 5.6	Table 5.2
II	Form workflow	Y	Two form workflows	5.6	Table 5.3
II	Scheduled workflow	Y	Defined; inactive in export (BUG-04)	5.6	Table 5.3
II	Approval process	Y	Leave_Approval	5.6	Fig. 5.13
II	Blueprint	Y	Appointment Management Blueprint	5.6	Fig. 5.12
II	Batch / report workflow	N	Not used	5.6	—
II	Testing & debugging evidence	Partial	Test plan written; execution results to be recorded	6	Table 6.1
III	Integration with a Zoho service	Y	Zoho Mail used only through sendmail	5.7	—
III	Connection / REST API call (invokeurl)	N	Not used	5.7	—
III	Webhook or Creator REST API	N	API access enabled in profiles; not used	5.7	—
III	Users, roles and permission profiles	Y	6 roles; permission profiles	5.8	Fig. 5.14

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Record-level access criteria / customer portal	Partial	Portal profiles defined; no record-level criteria	5.8	—
III	Zia / AI feature	Y	Not used	5.9	—
III	Analytics (charts / Zoho Analytics)	Y	Dashboard charts, pivot chart and table	5.9	Fig. 5.15
IV	Performance optimisation applied & measured	Partial	Applied; not benchmarked	7.2	Table 7.2
IV	Deployment across environments	Partial	[ENTER after publishing]	7.1	Fig. 7.1
IV	Security controls (encryption, OAuth, permissions)	Partial	Permissions and personal-data flags; no encrypted fields or OAuth	7.3	Table 7.3
IV	Governance (audit trail, administration)	Partial	[ENTER after capturing audit trail]	7.4	Fig. 7.2
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Justified in 5.10	5.10	—
V	Low-code limitations & spectrum analysis	Y	Sections 8.4 and 8.5	8.4, 8.5	—
—	Mobile app / portal access	Y	Phone and tablet layouts; portal profiles	5.11	Fig. 5.16

Total features fully implemented: 21 / 32 (partially implemented: 7; not implemented: 4)

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2		
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6		
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9		
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7		
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3		

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Patient_Profiles – Auto-generate Patient Code

```
// Script ID : DS-01
// Location  : Patient_Profiles → Form workflow → On Add → On Load
// Purpose   : Generate a unique patient code for a new
registration

if(input.ID == null || input.ID == 0)
{
    randomCode = randomNumber(100000,999999);
    currentDate = zoho.currentdate;
    dateStr = toString(currentDate,"yyyyMMdd");
    userEmail = zoho.loginuserid;
    if(userEmail.length() >= 3)
    {
        emailPrefix = userEmail.substring(0,3).toUpperCase();
    }
    else
    {
        emailPrefix = "PAT";
    }
    patientCode =
emailPrefix.concat(dateStr).concat(randomCode.toString());
    input.patient_code = patientCode;
}
```

C.2 Patient_Profiles – Cancel Duplicate Patient Entry

```
// Script ID : DS-02
// Location  : Patient_Profiles → Form workflow → On Add or Edit →
On Validate
// Purpose   : Reject a patient code that already exists

if(input.patient_code.isNull())
{
    // No patient code provided, allow submission
}
else
{
    existingPatients = Patient_Profiles[patient_code ==
input.patient_code];
    if(existingPatients.count() > 0)
    {
        alert "Error: Patient code " + input.patient_code + "
already exists in the system. Please use a unique patient code.";
        cancel submit;
    }
}
```

C.3 Appiontment_Scheduling – New Entry Notification

```
// Script ID : DS-03
// Location  : Appiontment_Scheduling → Form properties → Send e-
mail on success
// Purpose   : Notify the administrator of a new appointment entry
```

```
sendmail
[
    from :zoho.adminuserid
    to :zoho.adminuserid
    subject :"SmartCare - Clinic Management System : New entry
added"
    message :"Hello,<br/><br/>A new entry added in 'Appiontment
Scheduling' form of 'SmartCare - Clinic Management System'
application.<br/><br><br>"
]
```

C.4 Medical_Records – Follow-up Reminder

```
// Script ID : DS-04
// Location  : Medical_Records → Schedule → starts at Record_Date
(Asia/Kolkata)
// Purpose   : Remind the doctor to review the record (message
markup abbreviated)

sendmail
[
    from :zoho.adminuserid
    to :zoho.adminuserid
    subject :"Medical Record Follow-up Reminder"
    message :"<p>Dear Doctor,</p><p>This is a reminder to review
the patient's medical record and follow-up
requirements.</p><p>Please check the diagnosis, treatment plan, and
prescription, and take the necessary action.</p><p>Thank
you,<br>SmartCare - Clinic Management System</p>"
]
```

C.5 Apply_Leave – Approval Outcome E-mails

```
// Script ID : DS-05 / DS-06
// Location  : Apply_Leave → Leave_Approval → Level 1 → On Approve
/ On Reject
// Purpose   : E-mail the outcome to the applicant

// DS-05: on approve
sendmail
[
    from :zoho.loginuserid
    to :approver.submitter
    subject :"Approval Request"
    message :"Your request has been approved."
]

// DS-06: on reject
sendmail
[
    from :zoho.loginuserid
    to :approver.submitter
    subject :"Approval Request"
    message :"Your request has been rejected."
]
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

- Open the application link https://creatorapp.zoho.in/umarfarook.k.2028.cse/smartcare-clinic-management-system in a browser, or open the Zoho Creator mobile app and select SmartCare.
- Sign in with the user name and password supplied for your role. The left-hand menu shows only the sections your profile allows.

D.2 Instructions for the Receptionist

- Register a patient: open Patient Profiles → add a record. The patient code is filled in automatically; complete the other fields and click Submit. If a duplicate code alert appears, correct the code.
- Book an appointment: open Appiontment Scheduling → add a record, choose the patient and doctor, and click Submit. Open the record and use Confirm Appointment, Check In Patient or Cancel Appointment as the patient progresses.
- Raise an invoice: open Billing Invoices → add a record with the amounts and payment status.
- Apply for leave: open Leaves → Apply Leave, complete all fields and click Apply.

D.3 Instructions for the Doctor

- Open Appointment Calendar to see the day's appointments and use Start Consultation and Complete Consultation on the appointment record.
- Open Medical Records → add a record with diagnosis, treatment plan, prescription, visit type and follow-up details.
- Review patients in All Patient Profiles and past visits in All Medical Records or the Kanban Board.

D.4 Instructions for the Administrator

- Open Dashboard for total patients, total doctors and the distribution charts.
- Open Approvals → Pending Tasks to approve or reject leave requests; the applicant is notified by e-mail.

- Maintain staff in Staff Directory; use the Reports section for the pivot chart and pivot table; manage users, roles and profiles in the application settings.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	UMAR FAROOK K	Register No.	240701573
Project Title	SMARTCARE – CLINIC MANAGEMENT SYSTEM	Date	11-11-2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	UMAR FAROOK K	Register No.	240701573
Department / Section	CSE	Date of Viva	11-11-2026
Application URL Verified	Yes / No	Demo Duration	_____min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____